Benchmark report — CRC_FEQ_olive_O1_dualctx

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Model version: `0.1.0+CRC_FEQ_olive+65cfe77`
Checkpoint: `CRC_FEQ_olive_O1_dualctx.pt`
SHA-256: `451af695446ae0045d955566c311717b5570f6e6ff22b6d1376629bc6e5cbabd`
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Lab git (train): `65cfe77`
Code URL: https://github.com/catskillsresearch/fusion-equilibrium-challenge/commit/65cfe77
Phase-3 release asset: https://github.com/catskillsresearch/catskills_fec/releases/download/phase3-rebuild-6dc3637/CRC_FEQ_olive_O1_dualctx.pt
Not to confuse with: `CRC_FEQ_olive_geom` / O1G (later color; six-member ensemble parents). This report is **Olive O1 dualctx only**.

Problem description

What we are solving

Fusion Equilibrium Challenge (Challenge 1 — DIII-D): learn a mapping from *non-magnetic* tokamak diagnostics to the plasma's magnetic geometry.

In plain English: a tokamak is a donut-shaped bottle that holds hot plasma with magnetic fields. The shape of the plasma is a 2D map of **poloidal magnetic flux** $\psi(R, Z)$. Historically, **EFIT** (Equilibrium Fitting — the standard reconstruction code on DIII-D and many other tokamaks) rebuilds that map using magnetic sensors on the vessel wall. Future reactors will lose those sensors to radiation. We rebuild ψ (and two related numbers) from **coil currents**, **Thomson scattering** (T_e, n_e), and **plasma current** alone — with **no** Mirnov / pickup / Rogowski wall magnetics.

Ground truth is EFIT's labelled ψ on held-out DIII-D shots. The official scorer does **not** plug our $\hat{\psi}$ into the Grad–Shafranov PDE at scoring time; it

compares maps, boundaries, and geometry derived from the maps (details below). Grad–Shafranov is still the physics that EFIT solutions approximately satisfy, and the motivation for our physics-informed models.

Inputs (what we use)

All inputs are functions of time during a **shot** (one plasma pulse). Magnetics and Thomson are sampled at higher rate than EFIT; pipelines interpolate them onto EFIT timestamps t_k.

Symbol / name	Math (rough)	Plain English description
Coil currents	$I_c(t)$ for each coil c	How hard each electromagnet is driven. DIII-D: 18 F-coils (F1A…F9B), ohmic solenoid **ECOILA**, toroidal-field **bcoil** — shaping and heating "knobs."
Plasma current	$I_p(t)$	Total current flowing through the plasma (kA). Strong driver of equilibrium.
Thomson T_e	$T_e(t, \mathbf{x}_i)$	Electron **temperature** from laser scattering at several spatial channels $\mathbf{x}_i$. A **profile over time**, not one number.
Thomson n_e	$n_e(t, \mathbf{x}_i)$	Electron **density** on the same chords. Together $n_e T_e$ proxies kinetic pressure.
Core vs edge TS	two diagnostic views	Core vertical system vs edge/tangential system — different channel layouts.

Not used as challenge inputs (withheld): wall magnetic probes that classic EFIT relies on.

Outputs (what we submit / what we plot as Pred)

Output	Math	Plain English description
Flux map	$\hat{\psi}(R_j, Z_i;\ t_k)$ on a fixed	The full "topographic map" of magnetic surfaces. Nested closed contours = nested flux surfaces; outer closed

Output	Math	Plain English description
	65×65 (R, Z) grid per EFIT time	contour ≈ plasma edge.
Edge safety factor	$\hat{q}_{95}(t_k)$	**Edge safety** is how resistant the plasma edge is to certain MHD instabilities (higher $q \rightarrow$ field lines wind more helically, typically more stable against kink-like modes). q_{95} measures that winding number — times around the long way per short-way loop — on the ~95% flux surface (a standard near-edge surface; not the exact separatrix where $q \rightarrow \infty$). The range is about $[2, 23]$ on DIII-D shots here (most values ~3–9); **higher is "safer"** against kink-like modes. **We submit this as a scalar series.**
Normalized beta	$\hat{\beta}_N(t_k)$	Plasma pressure relative to magnetic pressure, normalized (Troyon-style β_N). The range is about $[-1.5, 4]$ on DIII-D shots here (most values ~0.2–2.6). **Higher means more pressure / performance** for given current and field, but too high approaches MHD stability limits — so "good" for fusion output is the high end, until the plasma hits those limits. **We submit this as a scalar series.**

Everything else the leaderboard scores is **derived by the scorer from our $\hat{\psi}$ —** we do not submit these numbers ourselves:

Derived quantity	Plain English description
LCFS outline	**Last closed flux surface** — the plasma's outer boundary curve in the (R, Z) plane.
Magnetic axis	The center of the nested flux surfaces — roughly where the plasma is "hottest / most central" in the cross-section (R_{axis}, Z_{axis}).
Elongation κ	How tall vs wide the plasma cross-section is (stretched up–down vs round).

Derived quantity	Plain English description
Triangularity	How D-shaped / pointed the cross-section is (inboard/outboard “corners”), not a simple circle or ellipse.
Volume	How much space the plasma occupies inside the LCFS.
Internal inductance ℓ_i	How peaked the toroidal current is toward the axis vs spread out — a shape-of-the-current profile number.

Scoring criteria (official composite S)

On each evaluation fold the scorer builds a single number in $[0, 1]$ (higher is better):

$$S = 0.55 \cdot \max(0, R^2_\psi) + 0.15 \cdot \max(0, R^2_{\{q_{95}, \beta_N\}}) + 0.10 \cdot \big(1 - \min(1, D_{\mathrm{LCFS}})\big) + 0.20 \cdot \mathrm{Cons}$$

Term	Weight	What it measures
R^2_ψ	0.55	Coefficient of determination between $\hat{\psi}$ and EFIT ψ over all grid points × times × shots. 1 = identical maps; ≤ 0 clipped to 0 in S.
$R^2_{\{q_{95}, \beta_N\}}$	0.15	Same idea for the two **submitted** scalars (pooled).
$1 - D_{\mathrm{LCFS}}$	0.10	Extract the **last closed flux surface** (plasma boundary polygon) from $\hat{\psi}$ and from true ψ; D_{LCFS} is a normalized symmetric **Hausdorff** distance between those curves. Closer outlines → higher score.
Consistency	0.20	From $\hat{\psi}$ (and from true ψ) compute geometric scalars: R_{axis}, Z_{axis}, elongation κ, triangularities, volume, ℓ_i. Score how well those **derived** numbers match — so a blurry-but-pretty ψ that implies the wrong shape is punished.

Physics equation (background, not the Codabench loss): ideal axisymmetric equilibrium obeys the **Grad–Shafranov** equation for poloidal flux $\psi(R, Z)$,

$$\Delta^* \psi = R \frac{\partial}{\partial R}\left(\frac{1}{R}\frac{\partial \psi}{\partial R}\right) + \frac{\partial^2 \psi}{\partial Z^2} = -\mu_0 R^2 p'(\psi) - F F'(\psi),$$

where $p(\psi)$ is plasma pressure and $F(\psi)$ relates to the toroidal field.
EFIT finds ψ consistent with that PDE plus measurements. Our official S above compares maps and geometry **directly**; a PINN-style trainer may still minimize a discretized Grad–Shafranov residual during learning so $\hat{\psi}$ stays physically plausible — that residual is **not** itself a term in S.

Olive model description — checkpoint `CRC_FEQ_olive_O1_dualctx`

Model ID: `CRC_FEQ_olive` *(public name; the word "olive" is only a color, not a method)*
Checkpoint tag: `O1_dualctx`
Color: Olive `#808000` — next luminance step after Steel Blue
Status (campaign): CLOSED 2026-08-20 as an early-SoftDerive color — SoftDerive never turned on; O1.1 never enabled; both parents died in data-only warmup. Successor color was Red.
Status (this checkpoint): RETAINED. `CRC_FEQ_olive_O1_dualctx.pt` is the from-scratch `CoeffDualContext` parent trained under Olive job **O1** (lab commit `65cfe77`). Official holdout at kill: $S = 0.986209$. Later `psi_tailfit` plain fine-tunes of this parent became the strongest single flux members in the house (COLOR_LEDGER §30.10); Phase 3 ships two of those children (`CRC_FEQ_brown_TF_LS_O1_{101,W1}`) and releases this parent as the warm-start asset.
What this report is: standalone Brown/Charcoal-style package report for **this specific weight**, integrating the Olive model description with the measured O1 receipt, post-campaign afterlife, and a 3-shot holdout figure pack under `figures/`. It is not a Codabench attempt directory.
Predecessor: `MODEL_DESCRIPTION_CRC_FEQ_steel.md` — closed.
Code: `pinn_charcoal.py` (`CoeffDualContext`, SoftDerive), `soft_derive.py`. GradNorm was new in this color.
Sibling (do not mix docs): olive_geom / O1G lives under `CRC_FEQ_olive_geom_*` — different MODEL_ID, different campaign.

This document is the model. It is written for a reader who does not already know fusion physics or neural networks. Every technical term is defined the first time it is used. The execution plan (which job to launch, and when) lives in `OLIVE_PLAN.md` and is not repeated here.

How to read this

The competition asks a computer to reconstruct the shape of a fusion plasma from sensor readings. Brown already does that as **two** networks whose predictions are averaged, and that mix is what we submit today. Steel tried to improve the mix by leaving those networks frozen and adding a small physics residual. That family is closed. Olive trains **new parents**: the same map family as B8 and F9, new weights, and a loss that can still pull κ and ℓ_i after pixel ψ has saturated.

The sections go in this order on purpose:

0. **The ledger** — every prior color, every process-step ID, goal, and measured result. Read this before proposing a job. It is the system prompt for what we already know.
1. The physical object we are reconstructing.
2. The three numbers the program must emit.
3. The formula the judges use to grade those numbers.
4. Why Steel is closed and why the leftover is fit quality.
5. The sensor readings and the 60-number picture.
6. Neural-network vocabulary, from zero.
7. Olive's two parents, drawn.
8. The **loss**, including GradNorm and the promoted ℓ_i stack.
9. The **optimizers**.
10. Every hyperparameter in one place.
11. When we mix (the three legal pairs, and the retired single-must-beat-0.99320 gate).
12. What we have already measured (Brown / Steel).
13. How to run it.

0. What prior colors already measured

This section is a **standing memory**, not a highlight reel. A later session that has not read Black through Steel should be able to start here and know which levers are dead, which ceilings are illegal, and which two bets Olive is allowed to take.

How to read the table. *Color* is the luminance revision. *ID* is the process-step tag on that color's board or plan (not a Codabench name). *Goal* is the one-variable question the step was written to answer. *Result* is the official 80-shot holdout unless marked public, oracle/GT, or existence-only. Verdicts:

word	meaning
SHIP	Packaged and scored on Codabench.
KEEP	Beats the then-held local bar. May or may not have shipped.
PAPER	Real but under the submit bar. Do not package.
PASS	The written gate cleared. Often a ceiling, not a submit.
MISS	Trained or applied; did not beat the keep bar.
KILL	Hit a written floor (usually R^2_ψ) and stopped.
CEILING	Uses holdout GT or an illegal warp. Not a submit.
LOCK	Negative result that closes a family.
NO	Existence fail; the arm was never legal.
MOVED	Rewritten as a later color's job.

Local S is the official scorer. Public S is Codabench Challenge 1 (DIII-D). Calibration has moved: maroon thought public ≈ local − 0.0013; purple ship was ≈ local − 0.0004; Brown B8+F9 is **public = local − 0.00135**. Do not mix those offsets.

0.1 One line per color

Color	Thesis	What it left
Black	Ridge / MLP on PCA-60	The family exists. Capacity is tiny.
Navy	Implicit neural $\psi(R, Z)$, no PCA	$S \approx 0.60$. INR as a parent is dead.
Teal	Train Consistency	Surrogate κ can go to $R^2 = -0.60$. Mean of three teals

Color	Thesis	What it left
	on the PCA manifold	still a toy.
Charcoal	Differentiable SoftDerive + geom emphasis	SoftDerive works. Geom-on MLP ships public 0.9556. Capacity, not the basis, is the fit limit. Vacuum-as-Y is dead.
Maroon	Full corpus + residual trunk + ensemble	Trunk > MLP. Clip 10. Concatenated context loses three times. R6: GT through $k = 60$ is Cons 0.9972. Public 0.9832 / #6.
Purple	Dualctx (context gets its own encoder)	C1/C2b/C7b. $\lambda_w = 1.35$ saturates. 200 epochs anti-align. GRU-only trunk dies. Public 0.9904 / #7.
Forest	Shot GRU + F9 + post-process κ	F9 single 0.99191. Oracles F15/F32/F57/F66 are ceilings. Predict s_Z, dR, dZ from X or the F9 trunk is dead.
Brown	Z-grid dualctx parent × F9	B8+F9 local 0.99320 → public **0.99185** / **#5**. Cousin 3-mixes dilute. B1/B3/B12/B15 kill.
Steel	Frozen-B8 residual (GS / probes / eddy)	Every arm ~0.991 or worse. K: $\Delta\kappa$ OOF $R^2 = -0.019$. Family closed.

Held ship is Brown B8+F9. Olive does not reopen any row marked LOCK / KILL / CEILING / NO.

0.2 The ledger

Color	ID	Goal	Result
Black	baseline	Ridge / small MLP on PCA-60	The competition family. S far below 0.95. Not the live ship.
Navy	INR	Drop PCA; emit $\psi(R, Z)$ from a coordinate MLP	**LOCK.** $S \approx 0.60$, Cons ~0. Do not rebuild the decoder as

Color	ID	Goal	Result
			an INR.
Teal	manifold Cons	Push Consistency in coefficient space	Surrogate can destroy κ ($R^2_\kappa = -0.60$). A biased Cons loss is worse than none.
Teal	ensemble3	Mean of three Wave-3 teal scale checkpoints	Packaging ID, not a luminance step. Toy vs later ships.
Charcoal	0a	Is $k = 60$ the Cons ceiling? GT through our basis	**PASS** / **LOCK.** Cons **0.9971**. FieldRefiner cancelled. Raising k buys nothing. (Maroon R6 repeats this cleaner.)
Charcoal	0b	Shrinkage vs noise: does smoothing still pay?	Ensemble and smoothing are substitutes. Smoothing pays only while incoherent residual remains.
Charcoal	0c	Test-time temporal smooth of ψ	Negative on a strong ensemble (−0.00035 S later on maroon R3). Helps slow scalars, hurts ℓ_i.
Charcoal	0d	Kappa error attribution	Field error, not extraction, dominates κ.
Charcoal	P1	Write SoftDerive = batched `derive_frame`	**PASS.** Discrete O-point / ψ_b detached; values stay in-graph. This is still the Cons loss.
Charcoal	P3	Ablate geom / temporal / H1	Geom emphasis works. Temporal window and H1

Color	ID	Goal	Result
			hurt at this scale.
Charcoal	P4	Scale s42, geom-emphasis 10	**KEEP.** Local S **0.9530** / Cons 0.8489. Beat the prior ship.
Charcoal	P5	Scale s202	**SHIP.** Local 0.9567 → public **0.9556**. Best charcoal. s101 crashed.
Charcoal	P6a	Soft D_{LCFS}	Term works as a surrogate; official $1 - D_{\mathrm{LCFS}}$ later shown not to track (maroon R10b).
Charcoal	P6b/c/d	Deficit weights / anneal / $L_{q\beta}$ as R^2	Built; ladder waited on P5. Not the maroon trunk.
Charcoal	P7	ψ diagnostic on trained-on vs holdout	**CAPACITY.** Trained-on R^2_ψ 0.99537. Not an input bug.
Charcoal	P8	Width/depth ladder at 250 shots	Width 512 helps a little; 1024+ overfits this shot count. Wrong question: we train on 12.7% of the corpus.
Charcoal	25.1	Subtract ψ_{vac} from Y	**LOCK.** Vacuum-as-Y dead. (Brown B13 later uses vac as *input* PCA, a different question.)
Maroon	M0	Coeff cache; reconstruct unseen shots	**PASS.** $R^2_\psi \geq 0.99999$. Full-corpus training fits on 12 GB.
Maroon	M1	Data ladder at equal steps, geom off	**PASS.** 7.7× data: S 0.92055 → 0.93443. Thesis holds. M2–M5 (capacity-first)

Color	ID	Goal	Result
			superseded.
Maroon	P1	Full corpus, geom-emphasis 10, 250 ep	**KILL** (power cycle ep 50). Partial S 0.95017: Cons 0.710→0.848, ψ and LCFS paid. Geom-on MLP is net-destructive on the term #1 wins.
Maroon	P2–P5	Continue charcoal geom ladder	**Cancelled.** P1 fragment was enough.
Maroon	R1	Residual trunk, geom off, vs M1c	**PASS.** S **0.97506**, R^2_ψ 0.99880. Trunk was the ceiling. Cons 0.8936 *without* geom loss.
Maroon	R2a	Concatenate window-5 context	**MISS.** S 0.95938. Central-diff of raw X amplifies jumps.
Maroon	R2b	`--feature-clip 10`	**KEEP.** S **0.97698**. Gain through Cons, not ψ. Clip stays on forever.
Maroon	R3	Seed ensemble of four R2b	**KEEP.** S **0.98170** / Cons 0.9231. Character ≫ magnitude (~80×). Seed variance ±0.002 S.
Maroon	R4	Context + clip together	**MISS.** S 0.96979. Concatenated context retired on this line.
Maroon	R5	17M trunk, no warmup / no clip	**VOID.** Diverged (loss → 7.9e6). Not a capacity verdict.

Color	ID	Goal	Result
Maroon	R6	GT ψ through $k = 60$	**LOCK.** S 0.99900 / Cons **0.9972** / κ **0.9887** / ℓ_i **0.9988**. The entire remaining gap is fit quality. Do not raise k.
Maroon	R7	R5 trunk + warmup 5 + grad-clip 1	**KEEP.** S **0.97998**. Capacity is real.
Maroon	R8	$1/N$ of seed members	Asymptote $S(N) = 0.98395 - 0.00910/N$. $N = \infty$ still under #1. Count is spent.
Maroon	R9	26.5M, matched wall clock vs R7	**MISS** as a lever. S 0.98002 (+0.00004). Capacity saturated.
Maroon	R10a	`smoothdiff` concatenated context	**MISS** as a single (S 0.97642) and **KEEP** as a mix member (+0.00035): different X representation.
Maroon	R10b	Soft LCFS on	**MISS** as a single. Official LCFS −0.00013 while the surrogate improved 17%. Proxy ≠ metric. Still adds +0.00019 in a mix.
Maroon	kopt	Is PCA-50 the missing ingredient?	14/60 modes lose to the mean; dropping them changes ψ MSE 0.001%. PCA-50 closed without training.
Maroon	ship6	Heterogeneous mix	**SHIP.** Local 0.98451 →

Color	ID	Goal	Result
		R7+R9+R2b+R3b+R10a+R10b	public **0.9832** / **#6**. Calib was wrong by 0.0045. Public ≈ local − 0.0013 *then*.
Purple	C1	Dualctx: context encoder + gate init 0	**PASS.** S **0.98178**. Cons +0.0085, ψ flat. Stem was the defect.
Purple	C2a	Bigger context encoder 1024×6	**MISS.** Below C1. κ −0.0157. Branch wants window, not width.
Purple	C2b	Window 5→9	Best single then: S **0.98214** / Cons 0.92619. Gate on S missed (±0.002). Cons is the signal.
Purple	combo	Free mix of owned ckpts	C1+C2a+C2b+R7+R9 local 0.98678. Not submitted (under then #5 bar).
Purple	C3	200 epochs of C2b	**MISS / LOCK schedule.** S 0.98154 *below* 88-ep C2b. ψ MSE down, Cons −0.00254. Objective anti-aligns past R^2_ψ 0.9993.
Purple	C4	8 dualctx members	Paused after C4a. Seed Cons noise is 0.0015, not 0.0005.
Purple	C5a	Whitened L_{coeff} clamp 12, $\lambda_w = 0.05$	**PASS.** S **0.98406** / Cons 0.93519. First Cons gain that does not spend ψ.
Purple	C5b	Val selector only	**LOCK selector.** ΔS −0.00008. Dead as a lever;

Color	ID	Goal	Result
			kept because it is cheap.
Purple	band	Hand true coeffs by mode band	Cons responds ~1:1 to gradient share. Modes 1–12 hold ~60% of recoverable Cons.
Purple	C6a	$\lambda_w = 0.15$, clamp 12	**MISS as written** (R^2_ψ −0.00005 vs gate) and **best single then**: S **0.98542** / Cons 0.94190. Dose curve is real.
Purple	C6b	Same strength, clamp 8	Allocation matters: lost 0.00326 Cons to C6a, won ψ .
Purple	C6c	Clamp 20 overshoot	Predicted miss; dropped to pay for C7.
Purple	C7a/b/c	λ_w 0.45 / **1.35** / 4.05	Lever saturates at **1.35**: S 0.98884 / Cons 0.95773. $\lambda = 4.05$ is a null (+0.00061 Cons).
Purple	C8a	Edge-weighted L_{data}	Used in the shipped 3-mix. Not a new #1.
Purple	C9a	Small geom on C7b	Cons pass / R^2_ψ miss. In the shipped 3-mix.
Purple	ship3	C7b+C8a+C9a	**SHIP.** Local 0.99096 → public **0.9904** / **#7**. Calib ≈ local − 0.0004 *then*.
Purple	D1a	GRU 2×256 as the only trunk	**KILL family.** S 0.96921 / Cons 0.867. R2a-shaped.

Color	ID	Goal	Result
Purple	D2	Gated GRU on frozen C7b	LIVE as a trainer, not a member. Every D2 mix lost to C7b+C8a+C9a. Shot recurrence → Forest.
Purple	396M GRU	GRU-only huge trunk	**KILL.** Val $R^2_\psi = -398$ after warmup.
Forest	F4	Shot GRU on frozen present-frame map	S 0.98919. Cousin: +0.00007 on the 3-mix. Not a new base.
Forest	F6	Unfreeze F4, lr 10^{-3}	**MISS.** S 0.98872.
Forest	F7	Unfreeze, lr 3×10^{-4}	**MISS.** S 0.98952.
Forest	F8	Unfreeze + C9a geom2	S 0.99014 / Cons 0.96254. Parent of F9.
Forest	F9	F8 + one geom-off data epoch (edge flag was a no-op)	**KEEP.** Best single **0.99191** / Cons 0.97037 / κ 0.9454. `best_epoch` 1. Held Forest single.
Forest	F10	F9 + edge band	**MISS.** 0.99171. Dose closed.
Forest	F11	F8 + edge mix 0.8	**MISS.** Tied F9 exactly.
Forest	F12	F9 + lcfs-emphasis 2	**MISS.** 0.99171 / Cons 0.9695. Soft LCFS does not move κ.
Forest	mix+F4	3-mix + F4	0.99103. Dilutes later F9.
Forest	F13	Inference GS projection of F9	**KILL.** S **0.861** / Cons 0.376. Do not project a finished map onto GS.
Forest	F14	Train-time L_{GS} 88 ep	**Never launched.** Flag was

Color	ID	Goal	Result
			not on the trainer. Same class as F13.
Forest	F15	Oracle s_Z vs GT κ	**CEILING.** S 0.99294 / Cons 0.9750. The knob exists.
Forest	F16	Features → s_Z	**LOCK.** corr 0.09; $s = 1.000 \pm 0.002$. S is F9.
Forest	F17	Features → κ then $s = \hat{\kappa}/\kappa_{\text{map}}$	**KILL.** Spent the map (κ 0.559).
Forest	F18	Map-shape → $s - 1$	**MISS.** corr 0.18. Stretch-from-anything-we-can-see closed.
Forest	F19	Hotter κ geom on finished F9, 4 ep	**KILL.** S 0.9823. Do not raise ε on a finished map.
Forest	F20	F15 + oracle dR	**CEILING.** S 0.99333 / R_{axis} 1.000.
Forest	F21	Features/shape → dR	**LOCK.** corr −0.02.
Forest	F22	F9 embed → dR	**MISS.** corr 0.12. Worse than F9.
Forest	F23	F8+edge+$R_{\text{axis}} \times 2$	**MISS.** S 0.9890. Walked off.
Forest	F24	5-frame MA of F9 ψ	**MISS.** S 0.99058.
Forest	F25	F9 recipe seed 101, 4 ep	**MISS.** S 0.9892. Val proxy picks a later epoch that spends the map.
Forest	F26	Seed 101, 1 ep	**MISS.** S 0.9859. F9 is a seed-42 ticket.
Forest	F27	Compute-light seq+mlp 512×2	**MISS.** Not F9.

Color	ID	Goal	Result
Forest	F28	Freeze F9, train only GRU	**MISS.** S 0.99138. Cousin of F4.
Forest	F29	C8a parent, F9 recipe 1 ep	**MISS.** S 0.9847.
Forest	F30	Oracle s_R for κ	**CEILING.** Width is not a better 1-dof than s_Z.
Forest	F31	F15 + s_R to restore volume	**MISS.** Next is isotropic scale.
Forest	F32	F15 + isotropic scale for volume	**CEILING.** PASS. GT only.
Forest	F33	Oracle dR on F32	**CEILING.** ψ spent at full dR.
Forest	F34	Scaled dR on F32	**CEILING.** Local #1 bar 0.9938 in sight.
Forest	F35	Legal shot-median κ stretch	**MISS.** Flattening kills frame variation.
Forest	F36	Iso-volume on raw F9	**MISS.** Volume returns; S does not. s_{iso} only works after s_Z.
Forest	F37	Legal MA5 κ stretch	**MISS.**
Forest	F38	dualseq 1 ep from F9	**KILL.** S 0.9837. Dropping the shot GRU walks off F9.
Forest	F39	F9 + L_{disp}	**MISS.** Held flux, spent Z_{axis} (0.886).
Forest	F40	High-mode coeff residual	**MISS.**
Forest	F41	F9 second epoch, lr 10^{-4}	**MISS.** S 0.9901. Any second step walks off.
Forest	F42	kNN stretch in F9 coeff space	**MISS.** Neighbors do not share s.

Color	ID	Goal	Result
Forest	F43	C9a + edge, stay dualctx	**KILL.** S 0.9761. Do not put the F9 recipe on another dualctx.
Forest	F44	q_{95}-implied κ stretch	**MISS.**
Forest	F45	F8 + edge-mix 0.4	**MISS.** S 0.9859.
Forest	F46	F8 + edge-power 2	**MISS** and a **bug**: dualshot `_loss` never applied `--edge-weight`. Bit-identical to F45. F9's gain is a geom-off data epoch, not edge.
Forest	F47	Real dualshot edge, hot lr	**MISS.** S 0.9873. 6× too hot vs F9's 5×10^{-5}.
Forest	F48	F8 + 1 ep, geom off, lr 5×10^{-5}	**MISS.** S 0.99044. F8 bump, not F9.
Forest	F49	F8 + 1 ep, lr 3×10^{-4}	**MISS.** Same basin as F47. F8+1-ep cannot reproduce F9.
Forest	F50	F9 + gentle real edge	**MISS.** S 0.99127. Spends F9 Cons.
Forest	F51	F9 + data-only second ep	**MISS.** Same as F50. Edge vs no-edge is a no-op. Second epoch closed.
Forest	F52	F8 + gentle edge	**MISS.** Same as F48.
Forest	F53	Shrink F9 κ to map mean	**KILL.** κ 0.842. Frame variation is real.
Forest	F54	F9+F4 mean	**MISS.** Dilutes F9 (−0.00083).

Color	ID	Goal	Result
Forest	F55	F8 + F9 4-ep schedule	**MISS.** Val picked epoch 4 and walked off.
Forest	F56	Embed → s_Z	**LOCK.** corr 0.059. Features, shape, coeffs, kNN, embed all fail to carry s.
Forest	F57	Oracle dZ on raw F9	**CEILING.** S 0.99267 / Z_{axis} 1.000. κ stays 0.945.
Forest	F58	Embed → dZ	**MISS** as a full apply (corr 0.36, first real signal). Full $\hat{dZ}$ is noise.
Forest	F59	$\alpha = 0.35$ embed-dZ	**PAPER.** S **0.99199** (+0.00008 vs F9). Not a ship (bar was 0.9924).
Forest	F60	Finer α	**LOCK.** Plateau 0.25–0.35. F59 stands.
Forest	F61	Ridge dZ at $\alpha = 0.35$	**MISS.** corr 0.13 < MLP 0.36.
Forest	F62	Embed+shape+features → dZ	**MISS.** Extra channels dilute (corr 0.09).
Forest	F63	F15 + oracle dZ	**CEILING.** S 0.99343. s_Z and dZ stack.
Forest	F64	F32 + oracle dZ	**CEILING.** S 0.99390. #1 stack starts here.
Forest	F65	kNN embed → dZ	**MISS.** corr 0.12.
Forest	F66	F32+dZ+$0.35\,dR$	**CEILING.** S **0.99436** / Cons 0.9830. Illegal GT warp. Olive's "match #1" local bar sits *above* this.

Color	ID	Goal	Result
Forest	F67	Features → dZ	**LOCK.** corr 0.06. Coils do not carry dZ.
Forest	F68	F8 + $Z_{\text{axis}} \times 3$	**MISS.** Bit-equal to F48. Z geom dose did not speak. Post-process family closed.
Brown	B1	Dualshot + stack geom from epoch 1, random init	**KILL.** $R^2_\psi = -228$ ep 1, still –48 at 30. Geom on a random field shatters ψ.
Brown	B2	Te-weighted kinetic Z apply on F9	**MISS.** S 0.983. F9 already has Z corr 0.984.
Brown	B2b	Affine leftover dZ from Thomson	**MISS.** leftover corr 0.08. Kinetic-Z post-process closed.
Brown	B3	Unfreeze C9a + stack after warmup	**KILL.** One Adam step, $R^2_\psi = -1.55$. Do not unfreeze finished F9/C9a.
Brown	B4	New dualctx `th1`, stack after 25	**MISS.** S 0.98992. Map-first on channel Thomson did not beat F9.
Brown	B5	Limiter-fill stretch of F9	**MISS.** S 0.970. Vessel-fill closed.
Brown	B6	Channel index vs Z (audit)	Layout is scrambled (Z std 55 mm). Why `thz` exists. Missed the `TS_edge` name.
Brown	B7	`thz` dualctx, $\varepsilon = 0.5$	**MISS** as single (0.99104). Do not 3-mix with B8.
Brown	B7+F9	Mean	KEEP then, superseded by B8+F9 (0.99298).

Color	ID	Goal	Result
Brown	B8	`thz` dualctx, $\varepsilon = 1.0$, warmup 25	**MISS** as single (**0.99173**). Owns Z_{axis} 0.9855. Do not raise ε.
Brown	B8+F9	Uniform mean	**SHIP.** Local **0.99320** / Cons 0.9754 / κ 0.9519 → public **0.99185** / **#5**. Held ship.
Brown	B8+B7+F9	3-mix cousins	**MISS.** 0.99306. Shared residual. Diluted.
Brown	B10	Thomson (R, Z) audit	**DONE.** `TS_tangential` is R-informative. B8 joined `TS_edge` (none).
Brown	B11	`thzr` dualctx	**MISS** single 0.99081; +F9 0.99283 (–0.00037 vs held); +B8 0.99224 (cousins).
Brown	B12	dualshot + `thzr`, geom after 25	**KILL.** S 0.0034 / $R^2_\psi = -3$. Same as B1.
Brown	B13	ψ_{vac} as 4 PCA *input* channels	**KEEP** vs B11 (0.99153). +F9 **0.99312** (–0.00008 vs held). Closest miss. Not a ship.
Brown	B14	(R, Z)+envelope CNN refiner	**MISS.** Cons unchanged vs B8.
Brown	B15	Trained free-boundary parent	**KILL.** Ep-8 $R^2_\psi = 0.982$. Holdout S 0.9794. Map formed, then spent.
Brown	A1	Cons autopsy of B8 / F9 / mix	**PASS.** Z is B8; R/ℓ_i are F9; κ shared ~0.945. A gate cannot invent κ.
Steel	A2	Learned scalar / 60-d gate on	**PAPER.** Scalar **0.99323**

Color	ID	Goal	Result
		frozen B8+F9	(+0.00003). Oracle w 0.99335. κ 0.952. Do not package.
Steel	S1	Frozen B8 + low-D (p', FF') + L_{GS} from ep 1	**KILL.** Holdout S **0.566** / Cons 0 / R^2_ψ 0.733.
Steel	S2	GS after warmup 8, $L_{\mathrm{GS}} = 0.005$	**MISS.** S **0.99172**. Best ep 2. Do not mix.
Steel	C0	Existence: poses, cited τ, shot join	**DONE.** C1 YES (GPEC poses). C2 YES (Sayer 6.9 / 3.3 ms). C3 **NO**.
Steel	C1-vac	Vacuum B/ψ PCA into residual head	**MISS.** S 0.99109 / κ 0.9406.
Steel	C1-reg	L_{mag} vs GT ψ at probes	**MISS.** S 0.99120 / κ 0.9399. (First build had `psi.detach()`; fixed; still theater.)
Steel	C1-shape	Same on κ / X-point subset	**MISS.** S 0.99077 / κ 0.9427. C1 closed.
Steel	C2-feat	Eddy PCA at $\tau = 6.9$ ms	**MISS.** S **0.99109** / κ 0.9429. Same class as C1-vac.
Steel	C2-rl	Predict eddy modes + RL loss	**MISS.** S **0.98971**. SoftDerive-on crashed Cons.
Steel	C2-tau33	Eddy PCA at $\tau = 3.3$ ms	**MISS.** S **0.99106**. τ is not the lever. C2 closed.
Steel	C3	Supervised current / `ECOILB` / shot hash	**NO.** No map, MDS private. Do not invent hashes or test `ECOILB`.

Color	ID	Goal	Result
Steel	C4	Longer shot context / GRU on frozen map	**LOCK** after K. `steel_ladder c4` exits 2.
Steel	K	$\Delta\kappa$ from coils / p_{th} / time / bilinear	**LOCK.** Mix OOF $R^2 = -0.019$. No Job 2. X already in B8/F9 does not explain κ.
Steel	SD	Add a second κ proxy	**LOCK.** B8 already has SoftDerive κ after ep 25. The hole is the map.
Steel	E1/E2	New parents (fit quality)	**MOVED** to Olive O1 / O2. Wrong color for a new parent.
Olive	P0	Wire GradNorm on h_{fuse}, λ_{geom} ramp, delayed w_g, ℓ_p floor	**NOT STARTED.** Blocks O1.
Olive	O1	New dualctx + GradNorm + ℓ_i -stack	**NOT STARTED.** Trainer not wired.
Olive	O1.1	PCGrad if O1 w_g collapsed	**CLOSED.** Never enabled. Collapse file is written only after geom is on and w_g dies.
Olive	O2	New dualshot from F8-class init, geom on after 8	**PARKED** until O1 holdout.
Olive	O3.a/b/c	Mix O1+O2 / O1+F9 / B8+O2	**PARKED** until holdouts.
Olive	P4	Next-color dimensionless-X if mixes miss	**NOT AN OLIVE JOB.**

0.3 Known knowns (laws, not hopes)

These are closed questions. Do not reopen them to be thorough.

1. **The basis is not the ceiling.** R6 / Charcoal 0a: GT through $k = 60$ is Cons 0.9972, κ

0.9887, ℓ_i 0.9988. Raising k, INR (Navy), FieldRefiner (0a), and X-point basis (Steel P) are dead.

2. **Pixel R^2_ψ is spent on the board.** We are 0.99964 public vs #1 0.99979. Consistency is ~74% of the weighted gap.
3. **A biased Cons surrogate is worse than none.** Teal κ; Charcoal/maroon geom-on MLP (P1); Soft LCFS (R10b, F12) moves the proxy and not the metric.
4. **Concatenated temporal context loses.** R2a, R4, R10a. Dualctx (C1) is the delivery that works. Gate starts at 0.
5. **Capacity saturated at ~17M residual / dualctx.** R7 vs R9: +56% params, +0.00004 S.
6. **Whitened L_{coeff} saturates at $\lambda_w = 1.35$.** C7c is a null. More λ will not buy the rest of Cons.
7. **Longer training can anti-align.** C3: 200 ep bought ψ MSE and spent κ. F9/F11/F12 restore `best_epoch` 1.
8. **Geom on a random field shatters ψ.** B1, B12. Warmup exists for this reason. Olive warmup is 8, not 0.
9. **Unfreezing a finished parent walks off.** B3 (F9/C9a), F19 (hotter ε on F9), F41/F50/F51 (any second F9 epoch).
10. **Post-process of F9 cannot see s_Z, dR, or a usable dZ.** F16/F21/F56/F58/F67. Oracles F15/F32/F57/F66 are GT ceilings, not submits. F66 0.99436 is the illegal local #1 stack.
11. **Cousin means dilute.** B8+B7+F9, B13+B8, F9+F4, 3-mix+F9. Heterogeneous residuals (B8 `thz` dualctx × F9 `th1` dualshot) are the mix that shipped.
12. **A gate on B8+F9 cannot invent κ.** A2 +0.00003; oracle +0.00015. A1: shared hole is $\kappa \sim 0.945$.
13. **Frozen-B8 residuals are spent.** S1/S2/C1/C2 all land at $S \approx 0.991$ or crash. Eddy τ 6.9 vs 3.3 is the same number.
14. **$\Delta\kappa$ is not in the legal coils.** K OOF $R^2 = -0.019$. No bilinear Job 2. No C4.
15. **We do not have test MPI / `EC0ILB` / a shot-hash map.** C3 is NO. Do not invent them.
16. **B8's stack floors ℓ_i.** Weights after mean-1: κ 2.561, ℓ_i **0.256**. F9 owns ℓ_i from a *geom-off* epoch. That split is already in the held mix.
17. **GRU as the only trunk dies.** D1a, 396M. Shot GRU as a *gated residual* on a map (F4/F9) is the form that lives.
18. **MAST is unsolved.** Public #1 0.73 vs us 0.055. Olive is a DIII-D parent hunt, not a transfer story.

0.4 Known unknowns (Olive is allowed to ask)

These are the only open questions that are not in the graveyard and that do not require a new illegal input.

#	Question	Why it is still open	How O1/O2 ask it
1	If SoftDerive turns on *while a new dualctx map is still plastic*, does κ move without spending R^2_ψ?	B8 waits until epoch 25; F9 never turns it on. F19/B3 change a *finished* map.	O1: data-only through epoch 8, then a 4-epoch λ_{geom} ramp. Kill if epoch-8 $R^2_\psi < 0.999$.
2	Does promoting ℓ_i off the floor write a different $\lvert\nabla\psi\rvert$ in the core?	B8 never paid ℓ_i. F9 got it for free, geom off. R6 says the basis *can* hold $\ell_i = 0.9988$.	O1 stack $\{\kappa : 1.5, Z, V, \ell_i : 1.0\}$. No third $\lVert\nabla\psi\rVert$ loss. ℓ_p floor 10^{-3}.
3	After R^2_ψ saturates, can GradNorm keep ∇L_{stack} alive?	Fixed ε either drowns the stack or, if raised on a finished map, spends ψ (F19). Navy Kendall-norm failed.	GradNorm $\alpha = 0.5$ on h_{fuse}, not on z. w_g Adam from epoch 10. PCGrad only if w_g collapses (O1.1).
4	Is a *new* dualshot with geom on better than finished F9?	F9 is one geom-off epoch on F8. B3 unfroze the *zip*. The unspent variable is geom on a dualshot that may still move.	O2, after O1 holdout, F8-class init, not `F9_unfreeze_edge`.

Expected if (1)–(3) work: $+0.0000$ to $+0.0006$ local S. Not a promise of 0.9945. A single is a mix member at $S \geq 0.99250$ (above either held parent), not only if it beats the mix. See §11.

0.5 Unknown unknowns (do not pretend we have a third parent)

If O1 and O2 both miss, the legal fit-quality lever is spent. What we do *not* know, and must not fill with a leftover Steel residual:

- Whether #1's Cons 0.9800 is a better *write* of $\hat{z}$ from the same X, or a legal input we still have not named. K says it is not the 21 coils, p_{th}, or time.
- Whether a third heterogeneous layout (not `th1`, not `thz`, not `thzr`) exists that is not B11/B13 theater. B10 closed `TS_edge`. Divertor has no Te/ne.
- Why their MAST S is 0.73. That may be the same unknown as the DIII-D Cons hole, or a different one. A model that scores 0.73 zero-shot on MAST is probably not regressing raw DIII-D coil amps onto EFIT pixels. A later color may *measure* non-dimensional X (scale by $I_p, B_{\mathrm{tor}}, a, R_0$). That is Phase 4 in `OLIVE_PLAN.md`, not an Olive job.
- Whether an organizer shot map or public `ECOILB` will ever appear. Until it does, C3 stays NO.

Do not invent an unknown unknown into a GPU job. The next legal move after two Olive misses is a new color with a new *measured* thesis, not another head on B8.

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1. The physical problem

A **tokamak** is a doughnut-shaped machine that holds a hot ionized gas (a **plasma**) with magnetic fields. The doughnut's horizontal radius is called R (meters from the machine's vertical axis). Height from the midplane is called Z (meters). A slice through the doughnut is therefore a 2-D (R, Z) plane.

One experimental run of the machine is a **shot** (also called a **discharge**). A shot lasts on the order of a few seconds. During the shot, a reconstruction code called **EFIT** writes down the plasma's magnetic geometry at a sequence of clock times. Each of those clock times is a **frame**. In this dataset a shot has between 1 and 445 frames; the middle (median) shot has 241 frames.

At each frame, EFIT stores a **poloidal flux map**. Flux is a scalar field: one number at every point in the (R, Z) plane. The conventional symbol is ψ (the Greek letter psi). The dataset stores ψ on a fixed **65 × 65 grid** — 4,225 numbers per frame. Those 4,225 numbers are the main thing we must predict.

Why ψ matters: the contours of ψ (the curves of constant flux) *are* the magnetic surfaces the plasma sits on. From those contours a physicist can read the plasma's boundary, its centre, how tall it is versus how wide, and several other shape numbers. The competition's "Consistency" score (defined in §3) is exactly "how well those derived shape numbers match the official ones."

We are **not** solving the Grad–Shafranov equation at apply time. Steel already tried a small current plus a Grad–Shafranov residual on frozen B8. It missed. We are doing **supervised learning**: given sensor readings X, emit a ψ that matches the EFIT ψ the laboratory already computed. The physics enters through (a) which sensors we are allowed to use and (b) how the official scorer — and our differentiable copy of it — reads geometry out of ψ.

The machine we train on is **DIII-D** (San Diego). Challenge 1 is "predict DIII-D equilibria the model has not seen." Challenge 2 (MAST, no training data) is packaged with the same code so the zip is complete. Public #1's MAST S is **0.73**; ours is **0.055**. Their method generalizes. Ours so far is a DIII-D-specific mix.

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2. What the program must emit

For every frame of every shot, a submission is three arrays:

name	shape	meaning
$\hat{\psi}$	65 × 65	predicted poloidal flux on the same grid as EFIT
$\hat{q}_{95}$	one number	**safety factor** near the edge. q is how many times a magnetic field line goes the long way around the doughnut per trip the short way. q_{95} is q evaluated at 95% of the way from the axis to the boundary.
$\hat{\beta}_N$	one number	**normalized beta**. Beta is plasma pressure divided by magnetic pressure. The N means a conventional normalization by plasma current and field so shots of different size can be compared.

Each parent also has three extra output heads (R of the magnetic axis, Z of the magnetic axis, and internal inductance ℓ_i). The official Challenge 1 scorer does **not** grade those heads. Olive's training loss (§8) does not use them either: it derives the same quantities from $\hat{\psi}$ the way the scorer does.

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3. How the judges score a prediction

3.1 Coefficient of determination, R^2

Given a list of true values y_i and predicted values $\hat{y}_i$,

$$R^2 \;=\; 1 \;-\; \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}$$

where $\bar{y}$ is the mean of the true values. $R^2 = 1$ is a perfect match. $R^2 = 0$ means the predictor is no better than always guessing the mean. R^2 can be **negative**: worse than guessing the mean.

When the y_i are all 4,225 pixels of every holdout frame, this number is called R^2_ψ. When they are the q_{95} and β_N time series, it is called $R^2_{q\beta}$.

3.2 The official composite S

The competition score is a weighted sum of four terms, each forced into $[0, 1]$:

$$S \;=\; 0.55\, R^2_\psi \;+\; 0.15\, R^2_{q_{95},\beta_N} \;+\; 0.10\,(1 - D_{\mathrm{LCFS}}) \;+\; 0.20\,\mathrm{Consistency}$$

- R^2_ψ (weight 0.55). Pixel-wise agreement of the flux map. Our models already sit at $R^2_\psi \approx 0.9996$. There is almost no score left to collect here.
- $R^2_{q_{95},\beta_N}$ (weight 0.15). Agreement of the two scalar time series.
- D_{LCFS} (weight 0.10). **LCFS** means *last closed flux surface*: the contour of ψ that is the plasma's boundary. D_{LCFS} is a **symmetric Hausdorff distance** (the largest gap between the predicted boundary polygon and the true one, in both directions), scaled so that 0 is perfect and 1 is "as bad as the scorer will count." The term in S is $1 - D_{\mathrm{LCFS}}$, so bigger is better.
- **Consistency** (weight 0.20). The mean of seven R^2 values, each computed on a shape number that the scorer *derives from* ψ:
  - **Magnetic-axis R, Z.** The (R, Z) location of the peak of ψ inside the plasma.
  - **Volume.** Cubic metres the plasma occupies, read from the boundary contour.
  - **Internal inductance ℓ_i.** How peaked the current is: high means concentrated near the axis.
  - **Elongation κ.** How tall the plasma is versus how wide.

- **Triangularity δ_{top}, δ_{bot}.** How pointed the top and bottom of the boundary are.

Public #1 is dmitryzhevn at S **0.99438** / Consistency **0.9800**. We are **0.99185** / **0.9707**. The top of the board is Consistency order. Pixel ψ is a ceiling.

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4. Why a new color

4.1 Steel is closed

Steel's thesis was: keep B8 frozen, predict a small current or a magnetics-shaped residual, and pull ψ toward physics the mix does not already encode. Every arm that trained landed in the same hole.

id	holdout S	Cons	κ	lesson
S1	0.566	0	—	L_{GS} from epoch 1 spent B8
S2	0.99172	0.9691	—	warmup kept the map; GS at 0.005 did not beat 0.99320
C1-vac / C1-reg / C1-shape	0.99077–0.99120	~0.969	0.940–0.943	vacuum PCA and probe regularizers are theater
C2-feat	0.99109	0.9686	0.9429	eddy extras at 6.9 ms = C1-vac
C2-rl	0.98971	0.9601	0.9318	RL eddy loss crashed Cons
C2-tau33	0.99106	0.9683	0.9420	3.3 ms = 6.9 ms. τ is not the lever
K	—	—	—	mix OOF $R^2(\Delta\kappa) = -0.019$. Lock. No Job 2. No C4

A learned gate on the *finished* mix (Steel A2) is +0.00003. The per-frame oracle gate is 0.99335. Kappa stays ~0.952. The cheap mix lever is spent.

4.2 The leftover is fit quality, not the decoder

Maroon R6 scored the $k = 60$ PCA reconstruction of *ground-truth* ψ:

	S	**Cons**	κ	ℓ_i
GT through our basis	0.99900	0.9972	0.9887	0.9988
Held B8+F9	0.99320	0.9754	0.9519	0.9692

The same linear eigenmodes that we decode through can hold κ at 0.9887. We sit at 0.9519 because the network does not *write* those coefficients, not because the basis cannot represent elongation. Raising k, switching to an implicit neural representation (Navy, $S \approx 0.60$), or a FieldRefiner (Charcoal 0a) is closed by that number.

The illegal Forest F66 warp of F9 (uses holdout ground truth) is local S **0.99436**. Submit bar is local **0.9945**. Olive does **not** promise that bar. Expected if O1 works: $+0.0000$ to $+0.0006$ local S.

4.3 What Olive is not

- Not a residual head on frozen B8 (Steel S / C).
- Not an unfreeze of finished F9 (Brown B3) or a hotter ε on finished B8 (Forest F19).
- Not Grad–Shafranov at inference (Forest F13, S 0.861).
- Not a new spatial decoder.

Olive spends the GPU on **who writes** $\hat{z}$ and **when SoftDerive is allowed to pull it**.

4.4 What the review added (accepted)

A later critique of this document (and of Steel's negative results) did not change the thesis. It changed four *delivery* details so O1 does not die of a known numerical failure mode, and it retired one mix gate that contradicted our own ship:

1. GradNorm's shared layer is h_{fuse}, not z (§8.4).

2. λ_{geom} ramps across epochs 9–12; w_g is frozen until epoch 10 (§8.2, §8.4).
3. ℓ_p in the ℓ_i denominator is floored at 10^{-3}; Huber 3σ stays (§8.2).
4. A single is a mix member at $S \geq 0.99250$, not only if it beats the held mix (§11). B8 and F9 never beat 0.99320 as singles.

PCGrad is O1.1, parked until w_g collapses. Non-dimensional X / the MAST 0.73 clue is Phase 4: a *next color*, not an Olive job. The job sequence is `OLIVE_PLAN.md`.

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5. Sensors and the 60-number picture

5.1 Legal inputs

Each frame is 149 raw numbers: 21 coil / shape currents and 128 Thomson-scattering channels (electron temperature and density). Challenge 1 forbids the magnetic probes EFIT used. Machine geometry (coil (R, Z), vessel outline) is an official input.

Thomson layout is how those 128 channels are ordered:

- `th1` — native channel order. This is F9.
- `thz` — the same measurements interpolated onto a **fixed height grid** in Z. This is B8, and it is O1.

The two layouts are different *coordinates* for the same sensors. That is why B8 owns Z_{axis} and F9 owns R_{axis} / ℓ_i.

5.2 PCA-60

A 65×65 map is 4,225 numbers. We store it as 60 **principal-component** coefficients plus a frozen mean field μ and basis C:

$$\psi \;=\; z\,C + \mu, \qquad z \in \mathbb{R}^{60}.$$

The network emits $\hat{z}$. Decode is a matrix multiply. The basis is frozen (B8's). Do not raise k.

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6. Neural-network vocabulary

A **neural network** is a stack of matrix multiplies and simple nonlinearities whose entries (the **weights**) are adjusted so that a **loss** gets small.

A **residual block** is two linear layers with the input added back: the block can learn "do nothing." **Width** is the length of the vector that travels through the block. **Depth** is how many such blocks are stacked. **Dropout** $p = 0.05$ randomly zeros 5% of coordinates during training.

A **GRU** (gated recurrent unit) reads a sequence one step at a time and keeps a hidden state. **Bidirectional** means a second GRU reads the shot backwards. F9 and O2 have a bidirectional GRU, hidden 256, one layer. B8 and O1 do not.

A **scalar gate** multiplies a context (or sequence) branch by one learned number before adding it to the present-frame embedding. The gate starts at **exactly 0**, so at step 0 the extra branch contributes nothing.

An **epoch** is one pass over every training shot. A **checkpoint** is a `.pt` file of the weights plus the feature μ, σ and the PCA basis. A **mix** is not a new network: run two finished checkpoints and average $\hat{\psi}$, $\hat{q}_{95}$, $\hat{\beta}_N$ in 64-bit.

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7. Olive's architecture

Olive is **one parent at a time**. O1 is a new dualctx. O2 is a new dualshot. They mix only after holdout, and only if the single is at least PAPER (§11). The held ship stays B8+F9 until a mix beats 0.99320.

flowchart TB shot["one discharge, T frames"] --> th1["th1 features
21 magnetics + 128 channel Thomson"] shot --> thz["thz features
21 magnetics + 128 Z-grid Thomson"] thz --> O1["O1 — new CoeffDualContext
19.27 M weights, random init
GradNorm on h_fuse + ℓ_i-stack after epoch 8"] th1 --> O2["O2 — new CoeffDualShot
from F8-class init, not F9
geom ON after epoch 8"] O1 --> p1["ψ̂, q95, βN from O1"] O2 --> p2["ψ̂, q95, βN from O2"] p1 --> gate{"single S ≥ 0.99250
and R²ψ ≥ 0.9993?"} p2 --> gate gate -->|KEEP-SINGLE or PAPER| mix["O3: O1+O2, O1+F9, B8+O2"] gate -->|MISS| hold["keep shipped B8+F9"]

O2 does not start until O1 has a holdout number. Do not start both on one GPU.

7.1 Temporal augmentation (both parents)

Let $x_0, x_1, \ldots, x_{T-1}$ be the raw 149-d frames of one shot. **Window 9** means a half-width of 4 frames. For each time t:

1. **Moving average** $\bar{x}_t$: the mean of $x_{t-4}, \ldots, x_{t+4}$, with indices clamped to $[0, T-1]$.
2. **Smooth difference** Δx_t: a central difference of the *average*,

$$\Delta x_t \;=\; \frac{\bar{x}_{t+4} - \bar{x}_{t-4}}{8}$$

The map's input is the concatenation $[x_t \mid \bar{x}_t \mid \Delta x_t]$, length 447. Windows never cross a shot boundary.

7.2 O1 — new `CoeffDualContext` (B8's trunk, new weights)

19,268,162 parameters. `--arch dualctx`. `--thomson-zgrid`. Random AdamW init. **Not** a load of `CRC_FEQ_brown_B8_zgrid_e1.pt`.

The two halves of the network never share a first layer. The present frame gets a wide residual tower. The 298 context numbers get a narrower tower. They meet as *embeddings*, not as raw columns, and the context tower is multiplied by the scalar `gate` which starts at 0.

flowchart TB subgraph inputs["one frame, 447 numbers — O1: thz"] raw["present raw x_t
149"] ctx["moving average + smooth difference
298"] end raw --> stem["Linear 149 → 1024"] stem --> act1["ReLU"] act1 --> t1["8 × ResBlock, width 1024"] t1 --> h["present embedding h
1024"] ctx --> cstem["Linear 298 → 512"] cstem --> act2["ReLU"] act2 --> cblocks["3 × ResBlock, width 512"] cblocks --> cproj["Linear 512 → 1024"] cproj --> gated["multiply by scalar gate
gate starts at 0"] h --> fuse["h_fuse = h + gated context
1024 — GradNorm shared layer"] gated --> fuse fuse --> heads["linear heads: z ∈ R60, q95, βN, R, Z, ℓ_i"] heads --> decode["ψ̂ = z C + μ"]

7.3 O2 — new `CoeffDualShot` (F8-class init, not F9)

20,418,627 parameters. `--arch dualshot`. Channel Thomson (`th1`). Init from an **F8-class** checkpoint, **not** `F9_unfreeze_edge`.

F9 was one **geometry-off** data epoch on F8. Brown B3 then tried to put stack geometry on the *finished* F9 zip and walked off in one Adam step. The unspent variable is geometry **on** a dualshot that is still allowed to move. That is O2. It waits on O1.

The map branch is the same dualctx tower as O1. A bidirectional GRU reads the raw 149-d frames of the whole shot and writes a per-frame residual into the embedding, multiplied by `gate_seq` (also init 0).

---

8. The loss

8.1 Flux terms (always on)

These four terms are B8's data loss. They run from epoch 1.

- L_{data} — ordinary mean squared error on the 65×65 ψ grid.
- L_{coeff} — mean squared error on the 60 raw coefficients. Coefficient 1 still dominates.
- $L_{\text{coeff,white}}$ — **whitened** coefficients. Let σ_k be the training-set standard deviation of coefficient k. Let $k_\star = 12$. The per-mode divisor is $s_k = \max(\sigma_k, \sigma_{k_\star})$. Then $L_{\text{coeff,white}}$ is MSE of $(\hat{z} - z)/s$. $\lambda_w = 1.35$. This is the purple C7 term that bought Consistency without a new decoder.
- $L_{q\beta}$ — MSE on q_{95} and β_N. $\lambda_{q\beta} = 1$.

Write $L_{\text{data+coeff}}$ for the sum of those four (with their usual coefficients). In GradNorm below, L_{data} means this bundle — the flux / scalar-head side — not the pixel term alone.

8.2 L_{stack} — SoftDerive after warmup 8

The official Consistency score is the mean of seven R^2 values the scorer *derives* from ψ. A biased surrogate of those numbers is worse than none (teal drove R^2_κ to -0.60).

Olive uses `soft_derive.py` — a batched, differentiable replica of `fusion_scoring.derive.derive_frame`. Discrete decisions (which cell is the O-point, which critical value is the boundary flux) use detached integer indices; every *value* read at those indices stays in the graph.

scalar	how SoftDerive computes it	why it is in the stack
κ	softmax over the co-area line measure $\lvert\nabla H\rvert$ of the LCFS; height / width	shared hole of B8 and F9 (~0.945)
Z_{axis}	parabolic sub-pixel fit about the strongest interior local maximum	B8 already owns this; keep it
volume	$2\pi \int R\, H\, dA$, exact as $\tau \to 0$	Forest s_{iso}
ℓ_i	volume average of B_p^2 over $(\mu_0 I_p/\ell_p)^2$	B8 left this on the floor; F9 owns it *without* geom

ℓ_i is a **volume** quadratic in $\nabla\psi$, not an LCFS extent:

$$\ell_i \;=\; \frac{\langle B_p^2\rangle_V}{(\mu_0 I_p/\ell_p)^2}, \qquad B_p = \frac{\lvert\nabla\psi\rvert}{R}, \qquad \langle B_p^2\rangle_V = \frac{\int B_p^2\, H\, dV}{\int H\, dV}.$$

PCA L^2 and even SoftDerive-κ (a boundary softmax) can match pixels and elongation while smearing $\lvert\nabla\psi\rvert$ in the core. Do **not** add $\lVert\nabla\hat{\psi} - \nabla\psi\rVert_2^2$ as a third loss. SoftDerive ℓ_i *is* that integral.

Let $g(\psi) \in \mathbb{R}^7$ be SoftDerive on a frame, in the order `R_axis, Z_axis, kappa, tri_top, tri_bot, volume, li`.
Let σ_j be the training-set standard deviation of reference scalar j. The per-scalar residual is Huberised at 3 σ:

$$r_{t,j} = \frac{g_j(\hat{\psi}_t) - g_j(\psi_t)}{\sigma_j}, \qquad \rho(r) = \begin{cases} r^2 & \lvert r\rvert \le 3 \\ 3\,(2\lvert r\rvert - 3) & \lvert r\rvert > 3 \end{cases}$$

Do not turn SoftDerive on at epoch 1 from random weights (Brown B1 shattered ψ; R^2_ψ still -48 at epoch 30). Warmup is **8** epochs of $L_{\text{data+coeff}}$ only — shorter than B8's 25 because we already know this trunk forms a map, and we want the stack on while the map is still plastic. Kill if epoch-8 val $R^2_\psi < 0.999$ or val Cons < 0.96. Epoch 8 is the last data-only epoch: the kill is a health check, not a post-shock reading.

Do **not** step L_{stack} on at full λ_{geom} on epoch 9. That shocks AdamW's momentum buffers and can crash R^2_ψ just as the kill window closes. The multiplier on λ_{geom} is

epoch	≤ 8	9	10	11	12	≥ 13
multiplier	0	0.25	0.50	0.75	1.0	1.0

This is separate from the **learning-rate** warmup (5 epochs, §9). Do not conflate them.

ℓ_i's denominator uses the poloidal perimeter ℓ_p. SoftDerive today clamps ℓ_p at `eps = 1e-12`. A vanishing contour during early geom-on frames spikes $\nabla \ell_i$. Olive's trainer must use $\ell_p \leftarrow \max(\ell_p, 10^{-3})$ in that denominator only (a flag; do not silently change charcoal / Brown). The Huber 3σ threshold stays strictly on.

8.3 Promoted ℓ_i stack

B8's stack (`STACK_GEOM_FOCUS = {kappa: 1.5, Z_axis: 1.0, volume: 1.0}` , floor 0.15) puts ℓ_i on the floor. After normalising to mean 1:

	R_{axis}	Z_{axis}	κ	δ_{top}	δ_{bot}	**volume**	ℓ_i
B8 (today)	0.256	1.707	2.561	0.256	0.256	1.707	**0.256**
Olive	0.193	1.284	1.927	0.193	0.193	1.284	**1.284**

Olive sets `STACK_GEOM_FOCUS` to $\{\kappa : 1.5,\ Z : 1.0,\ V : 1.0,\ \ell_i : 1.0\}$, same floor 0.15. κ stays the heaviest term. ℓ_i is promoted to the same mass as Z and volume. Huber and the scorer's plausibility mask stay.

$$L_{\text{stack}} \;=\; \frac{\sum_{t,j} \mathbf{1}_{t,j}\, w_j\, \rho(r_{t,j})}{\sum_{t,j} \mathbf{1}_{t,j}\, w_j}$$

The geometry term is already dimensionless. The *emphasis* that puts it on the same scale as L_{data} is still

$$\lambda_{\text{geom}} \;=\; \varepsilon \cdot \frac{0.20}{0.55} \cdot \text{Var}(\psi)$$

with $\varepsilon = 1.0$. That scalar is **not** the adaptive weight. GradNorm is.

8.4 GradNorm

After warmup, two losses fight for the same weights. Once R^2_ψ is 0.9996, ∇L_{data} is small in *score* units and still large in *parameter* units; it drowns $\nabla\kappa$ and $\nabla\ell_i$. Raising ε on a finished map (F19) spends ψ. GradNorm (Chen, Zhao, Xu, Koltun, ICML 2018) trains a **scalar** $w_g(t)$ so the two gradient norms stay in a chosen ratio. $w_\psi = 1$ is fixed.

The shared layer θ is the **fused embedding** $h_{\text{fuse}} \in \mathbb{R}^{1024}$ immediately before the linear heads (`fuse` in §7.2; for O2, after the `gate_seq` residual is added). Do **not** take ∇_θ through the PCA head $z \in \mathbb{R}^{60}$. That would let GradNorm see only the last linear layer and miss the trunk.

Let $G_\psi = \|\nabla_{h_{\text{fuse}}} L_{\text{data+coeff}}\|_2$ and $G_g = \|\nabla_{h_{\text{fuse}}}(w_g L_{\text{stack}})\|_2$. Let $\bar{G}$ be their mean. The relative inverse training rate of the stack is

$$r_g(t) \;=\; \frac{L_{\text{stack}}(t)/L_{\text{stack}}(0)}{L_{\text{data+coeff}}(t)/L_{\text{data+coeff}}(0)}.$$

GradNorm's target is $G_g \to \bar{G}\cdot(r_g)^\alpha$ with $\alpha = 0.5$. w_g is updated by a small Adam step on $|G_g - \text{stopgrad}(\bar{G}\cdot r_g^\alpha)|$, then renormalised so the two task weights still sum to 2. The network weights see

$$L \;=\; L_{\text{data+coeff}} + w_g\,\lambda_{\text{geom}}\,L_{\text{stack}}.$$

Initialise $w_\psi = 1,\ w_g = 1$ at the first geom epoch. Do **not** Adam-update w_g until **epoch 10** (the second ramp epoch). Epochs 9–10 run the stack at unit GradNorm weights so the gradient ratio is measurable before the weight starts chasing it.

Navy's `--loss-ema-norm` (Kendall uncertainty) failed. Use GradNorm first. **PCGrad** (project each gradient off the other when their dot product is negative) is the backup if GradNorm's w_g collapses ($< 10^{-3}$) or explodes (> 5) and Cons is flat; that job is **O1.1**, not O1.

Detached integer O-point / ψ_b in SoftDerive already keeps $\nabla\kappa$ and $\nabla\ell_i$ defined almost everywhere.

8.5 Trainer contracts (Phase 0)

These four must be in code before O1 launches. They are the accepted engineering refinements of the Olive thesis, not new hypotheses.

id	contract
P0.a	GradNorm ∇ is at h_{fuse}, not at z.
P0.b	λ_{geom} ramp: 0 through epoch 8, then $0.25/0.50/0.75/1.0$ on epochs 9–12.
P0.c	w_g Adam starts at epoch 10. Unit weights until then.
P0.d	ℓ_i denominator uses $\ell_p \leftarrow \max(\ell_p, 10^{-3})$. Huber 3σ stays.

---

9. Optimiser and schedule

name	value
Optimiser	AdamW, weight decay 10^{-4}
Learning rate	3×10^{-4} (slower than B8's 10^{-3}; SoftDerive is on at epoch 8)
Warmup	5 epochs, linear, then cosine to 0
Epochs	88
Grad clip	1.0
L-BFGS	off
Train shots	6,961
Official holdout	80 shots, `--seed 42`
Feature clip	10 σ
SoftDerive warmup	8 (data-only through epoch 8)
λ_{geom} ramp	epochs 9–12: $0.25, 0.50, 0.75, 1.0$
GradNorm α	0.5; ∇ at h_{fuse}; w_g Adam from epoch 10
ℓ_p floor	10^{-3} in the ℓ_i denominator
Kill	epoch-8 val $R^2_\psi < 0.999$ or val Cons < 0.96

The number we *report* is always the official scorer on the 80-shot holdout, run after training finishes.

---

10. Hyperparameters, all in one place

10.1 Data

name	O1	O2
Train shots	6,961	6,961
Official holdout	80, seed 42	80, seed 42
Raw features	149 (`thz`)	149 (`th1`)
Dual features	447	447
PCA modes	60, frozen (B8 basis)	60, frozen
Feature clip	10	10

10.2 Present-frame map (both)

name	value
Stem	Linear 149 → 1024
Trunk width × blocks	1024 × 8 residual blocks
Context stem	Linear 298 → 512
Context width × blocks	512 × 3 residual blocks
Context projection	Linear 512 → 1024
`gate`	1 scalar, init 0
Activation	ReLU

name	value
Dropout	0.05
Map parameters	19,268,162

10.3 What differs

name	O1	O2
Class	`CoeffDualContext`	`CoeffDualShot`
Shot GRU	none	bidirectional, hidden 256, 1 layer
`gate_seq`	—	1 scalar, init 0
Total parameters	19,268,162	20,418,627
Thomson	Z-grid `thz`	channel `th1`
Init	random AdamW	F8-class checkpoint, **not** F9
ε, warmup	1.0, 8	1.0, 8
Stack	κ, Z, V, ℓ_i	same
λ_{geom} ramp	epochs 9–12	same
GradNorm	$\alpha = 0.5$ on h_{fuse}; w_g from ep 10	same
When	first (after P0)	after O1 holdout

---

11. Mixing

A **mix** is: run two finished checkpoints on the same shot, then average $\hat{\psi}$, $\hat{q}_{95}$, and $\hat{\beta}_N$ in 64-bit. No new training.

B8 as a single is 0.99173. F9 as a single is 0.99191. The held product is their mean at 0.99320. Requiring a *new* single to beat 0.99320 before mixing would have rejected both parents of the

ship. That gate is **retired**.

verdict	single S	then
KEEP-SINGLE	≥ 0.99250, Cons ≥ 0.9720, $\kappa \geq 0.9480$, $R^2_\psi \geq 0.9993$	Mix member.
PAPER	$0.99173 < S < 0.99250$ and Cons or κ beat the matching held parent	Different character. Mix is allowed.
MISS	≤ 0.99173 and no Cons/κ gain	Not a mix member.

id	mix	when
O3.a	$0.5\,\mathrm{O1} + 0.5\,\mathrm{O2}$	both at least PAPER
O3.b	$0.5\,\mathrm{O1} + 0.5\,\mathrm{F9}$	O1 at least PAPER
O3.c	$0.5\,\mathrm{B8} + 0.5\,\mathrm{O2}$	O2 at least PAPER

A mix $S > 0.99320$ replaces the held *local*. Package only at ≥ 0.99350. Submit only at ≥ 0.9945. Prefer a mix that also moves $\kappa > 0.9519$ or Cons > 0.9754.

Do not 3-mix B13 or B7. Dualctx cousins share a residual. Do not A2-gate B8+F9 again. Do not add a MISS as a third member.

---

12. What we have already measured

All local numbers are the official scorer on the same 80-shot holdout.

artifact	local S	Cons	R^2_ψ	κ	what it is
F9 single	0.99191	0.9704	0.99945	0.9454	best *single*; `th1` dualshot, geom off
B8 single	0.99173	0.9692	0.99961	0.9445	`thz` dualctx; owns Z_{axis} 0.9855

artifact	local S	Cons	R^2_ψ	κ	what it is
B8+F9 (held, public #5)	**0.99320**	**0.9754**	**0.99966**	**0.9519**	uniform mean
A2 oracle w	0.99335	0.9757	0.99980	0.9515	per-frame MSE-optimal mix; GT on holdout
A2 scalar gate	0.99323	0.9755	0.99970	0.9521	paper keep; do not package
C2-feat / C2-tau33	0.99109 / 0.99106	0.9686 / 0.9683	0.99956	0.943 / 0.942	frozen-B8 eddy; closed
C2-rl	0.98971	0.9601	0.99961	0.9318	worse; closed
F66 oracle stack	0.99436	0.9830	0.99930	—	illegal GT warp of F9; ceiling, not a submit
R6 GT through $k = 60$	0.99900	0.9972	—	0.9887	basis is not the ceiling
Public #1 dmitryzhevn	0.99438*	0.9800*	0.99979*	—	*public*; MAST S 0.73 vs our 0.055

A1 autopsy (same 80 shots, no re-infer):

- **Disagree** ($|\Delta| \geq 0.005$): R_{axis} F9 +0.0085, Z_{axis} B8 +0.0165, ℓ_i F9 +0.0098.
- **Shared hole:** κ (B8 0.9445 / F9 0.9454). A gate can pick Z vs R; it cannot invent elongation.

Public board 2026-08-20 07:52 (Codabench 19740), Challenge 1 DIII-D:

rk	who	S	Cons	R^2_ψ
1	dmitryzhevn	0.99438	0.98002	0.99979
2	dgaytanv	0.99320	0.97608	**0.99986**
3	mnn31	0.99208	0.97374	0.99977

rk	who	S	Cons	R^2_ψ
4	mwaller	0.99196	0.97049	0.99972
5	**catskills**	**0.99185**	**0.97072**	**0.99964**

Weighted gap #5 vs #1 (0.00253 of S): Cons 74%, LCFS 13%, $q\beta$ 10%, ψ 3%.

Closed — do not reopen

- Steel S1 / S2 / C1 / C2 / C3 / C4 / K / A2-as-ship.
- F9 post-process (s_Z, dR, dZ, fill).
- Raise ε on finished B8 (F19).
- Unfreeze finished F9 / C9a (B3).
- Inference GS (F13, S 0.861).
- Vacuum-as-Y (Charcoal 25.1).
- Raise k / INR / FieldRefiner / X-point basis (R6 / Navy / Charcoal 0a).
- 3-mix of dualctx cousins (B7, B13).
- Stack geometry from random dualshot weights (B1).

12b. What this checkpoint actually scored (O1 dualctx)

Olive's plan doc (§12 above) was written **before** O1 ran. The receipt below is the official scorer on the same 80-shot holdout after training stopped.

Kill reason. SoftDerive warmup was 8 epochs. Epoch-8 health gate required val $R^2_\psi \geq 0.999$ (and Cons ≥ 0.96). Training selected `best_epoch=7` still in the data-only ramp; SoftDerive / GradNorm **never turned on**. Tag `O1_dualctx`. Arch `dualctx`. Seed / init_seed 42.

Term	Value
S	**0.986209**

Term	Value
R^2_ψ	0.998504
$R^2_{q\beta}$	0.989851
$1 - D_{\mathrm{LCFS}}$	0.983053
Consistency	0.951246

Per-scalar Consistency R^2 (fail frac all 0.0):

Scalar	R^2
R_{axis}	0.950738
Z_{axis}	0.951980
κ	0.920411
δ_{top}	0.976302
δ_{bot}	0.961494
volume	0.953805
ℓ_i	0.943993

Verdict at the time: KILL (campaign). Holdout S was well below B8's 0.99173; the map was not yet championship-healthy. The checkpoint was kept on disk.

O1.b (adaptive warmup from O1 epoch 7) — separate run, not this weight: holdout $S = 0.989887$, Cons 0.962833, $R^2_\psi = 0.999235$, also **KILL**. SoftDerive still never delivered the Olive thesis.

Receipts: lab `results/olive/O1_dualctx.json`, `results/CRC_FEQ_olive_O1_dualctx_score.json` (byte-identical scores).

13. How to run (historical) and how this weight is used now

Historical (Olive campaign, lab `65cfe77`). Env `CRC_FEQ_MODEL_ID=CRC_FEQ_olive`. Board was `:8771`. Ladder entry:

```
CRC_FEQ_MODEL_ID=CRC_FEQ_olive python -m catskills.scripts.olive_ladder o1
```

That produced `artifacts/CRC_FEQ_olive_O1_dualctx.pt` (SHA-256 above). Do not re-run expecting SoftDerive — the campaign closed with SoftDerive never on.

Phase 3 / organizer rebuild (this repo). Place the release weight where `psi_tailfit` expects the O1 parent (`artifacts/` or `CRC_FEQ_WEIGHTS_DIR`), then:

```
python -m catskills.scripts.psi_tailfit --parent O1 --power 0.0 --seed 101 \
  --tag LS_O1_101 --epochs 220 --patience 30
# → CRC_FEQ_brown_TF_LS_O1_101.pt

python -m catskills.scripts.psi_tailfit --parent O1 --power 0.0 --seed 42 \
  --lambda-w 1 --tag LS_O1_W1 --epochs 220 --patience 30
# → CRC_FEQ_brown_TF_LS_O1_W1.pt
```

See `ORGANIZER_REPLY_REBUILD.md` §1.5 and `MANIFEST.md`.

---

14. Artifacts and code map

This weight	`weights/phase3/CRC_FEQ_olive_O1_dualctx.pt` (also lab `archive/artifacts/…`, same SHA-256)
This report	`weights/phase3/REPORT_CRC_FEQ_olive_O1_dualctx.md`
Model description (source)	`MODEL_DESCRIPTION_CRC_FEQ_olive.md`

Score receipt	lab `results/olive/O1_dualctx.json`
Tailfit children (Phase 3)	`CRC_FEQ_brown_TF_LS_O1_101.pt`, `CRC_FEQ_brown_TF_LS_O1_W1.pt`
Tailfit receipts	lab `results/brown/P30_TAILFIT_LS_O1_{101,W1}.json`
Env (train era)	`CRC_FEQ_MODEL_ID=CRC_FEQ_olive`
Not this model	`CRC_FEQ_olive_geom_*` / O1G

file	what it is
`pinn_charcoal.py`	`CoeffDualContext`, stack geom, `predict_row`
`soft_derive.py`	differentiable seven scalars (never engaged on this checkpoint)
`scripts/psi_tailfit.py`	plain / weighted fine-tune that resurrected O1
`COLOR_LEDGER.md` §30.10	undertrained-parents finding; O1 → best single

---

15. Afterlife — why this “killed” weight still matters

COLOR_LEDGER §30.10 (*The parents were undertrained*) re-measured holdout ψ MSE under plain `psi_tailfit` ($p = 0$, coefficient MSE, frozen PCA-60). **O1 was the worst member on the §30.7 board by rank, then the best model in the project after a proper budget:**

stage	holdout ψ MSE	note
O1 parent (this weight), as left by Olive	3.78×10^{-3}	undertrained; SoftDerive never on
Short plain fine-tune (O1/U)	5.9115×10^{-4}	**–84%** vs parent

stage	holdout ψ MSE	note
Long plain fine-tune (O1/L, 220 ep)	5.9275×10^{-4}	same class

Best early mix including O1 children: B8/L + O1/U + O1/L → 5.2410×10^{-4} (**+38.40%** cut vs B8+F9 pair MSE). Hybrid with locked B8+F9 $q\beta$: local $S = \mathbf{0.994147}$ (public calibration 916155: $d3d_S = 0.993112$).

Phase 3 members descended from this parent (receipts, not re-scored here):

child	holdout ψ MSE	cut vs B8+F9 pair (8.508×10^{-4})	best_epoch
`LS_O1_101`	5.361×10^{-4}	37.0%	90
`LS_O1_W1` ($\lambda_w = 1$)	6.025×10^{-4}	29.2%	142

So the Olive *thesis* (early SoftDerive + GradNorm) failed, but the Olive *trunk* (new dualctx, B8 recipe, `thz`, stack geom scheduled) left a plastic parent that later campaigns mined. This report documents that parent.

Explicitly out of scope: `CRC_FEQ_olive_geom` overnight master blends, O1G lp-shift seeds, championship packages under `benchmarks/archive/CRC_FEQ_olive_geom/` — those are a later color with sketchier per-attempt `REPORT.md` stubs. Do not treat them as this weight's benchmark.

Scores (official composite) — this checkpoint

Official `fusion_scoring` metric **3.1.0**, machine **DIII-D**, 80-shot holdout, seed 42.

Term	Value
S	0.986209

Term	Value
R^2_ψ	0.998504
R^2_qβ	0.989851
1−D_LCFS	0.983053
Consistency	0.951246

Per-scalar Consistency R^2 (unclipped; fail frac all 0):

Scalar	R^2	fail frac
R_axis	0.950738	0.000
Z_axis	0.951980	0.000
kappa	0.920411	0.000
tri_top	0.976302	0.000
tri_bot	0.961494	0.000
volume	0.953805	0.000
li	0.943993	0.000

Training Parameters

Parameter	Value
method_name	CRC_FEQ_olive
tag	O1_dualctx
arch	dualctx
model_version	0.1.0+CRC_FEQ_olive+65cfe77
train shots	6961

Parameter	Value
test shots (n_eval)	80
PCA components	60
temporal_window	9
thomson_layout	zgrid (`thz`)
optimizer	adamw
epochs (scheduled)	88
best_epoch	7
best_selection	0.731276
seed / init_seed	42 / 42
lambda_geom	0.014477094466036016
geom_emphasis	1.0
geom_warmup	8
geom_weights	stack (Olive promoted ℓ_i)
geom_weight_vec	R_axis 0.212, Z_axis 1.414, kappa 2.121, tri_top 0.212, tri_bot 0.212, volume 1.414, li 1.414
lambda_coeff_white	1.35
coeff_white_clamp	12
lambda_lcfs	0
edge_weight	off
lr_warmup	5
grad_clip	1.0
val_fraction	0.1 (shot split)

Parameter	Value
val_geom_frames	20000
SoftDerive engaged?	**no** (killed in data-only warmup)
GradNorm engaged?	**no**
train_s	606.5
infer_holdout_s	17.7
checkpoint	`CRC_FEQ_olive_01_dualctx.pt`
git_commit (train)	65cfe77

Qualitative summary

Shot 0 (`d3d_shot_ea67ad907d`)

- Flux MSE is low (2.69e-05) — pixel-level agreement looks strong on this shot.
- Error is fairly balanced between edge and core (ratio 1.40).
- Amplitude scale factor ≈ 1.006 — overall gain looks reasonable.
- Largest |Diff| at mid-time near R=2.54 m, Z=-0.45 m (|err|=0.00932).
- q95: RMSE=0.189 vs EFIT — the side-model that predicts this submitted number (separate from the 65×65 flux path) is missing the target; that hurts the 0.15 weight on $R^2_{q95,\beta N}$ in S.
- betaN: RMSE=0.0456 vs EFIT — the side-model that predicts this submitted number (separate from the 65×65 flux path) is missing the target; that hurts the 0.15 weight on $R^2_{q95,\beta N}$ in S.

Shot 1 (`d3d_shot_c965ecee3a`)

- Flux MSE is low (4.74e-05) — pixel-level agreement looks strong on this shot.
- Error is fairly balanced between edge and core (ratio 0.87).
- Amplitude scale factor ≈ 0.983 — overall gain looks reasonable.
- Largest |Diff| at mid-time near R=2.54 m, Z=0.50 m (|err|=0.00871).

- q95: RMSE=0.185 vs EFIT — the side-model that predicts this submitted number (separate from the 65×65 flux path) is missing the target; that hurts the 0.15 weight on $R^2_{q95,\beta N}$ in S.
- betaN: RMSE=0.0562 vs EFIT — the side-model that predicts this submitted number (separate from the flux-map path) matches well on this shot.

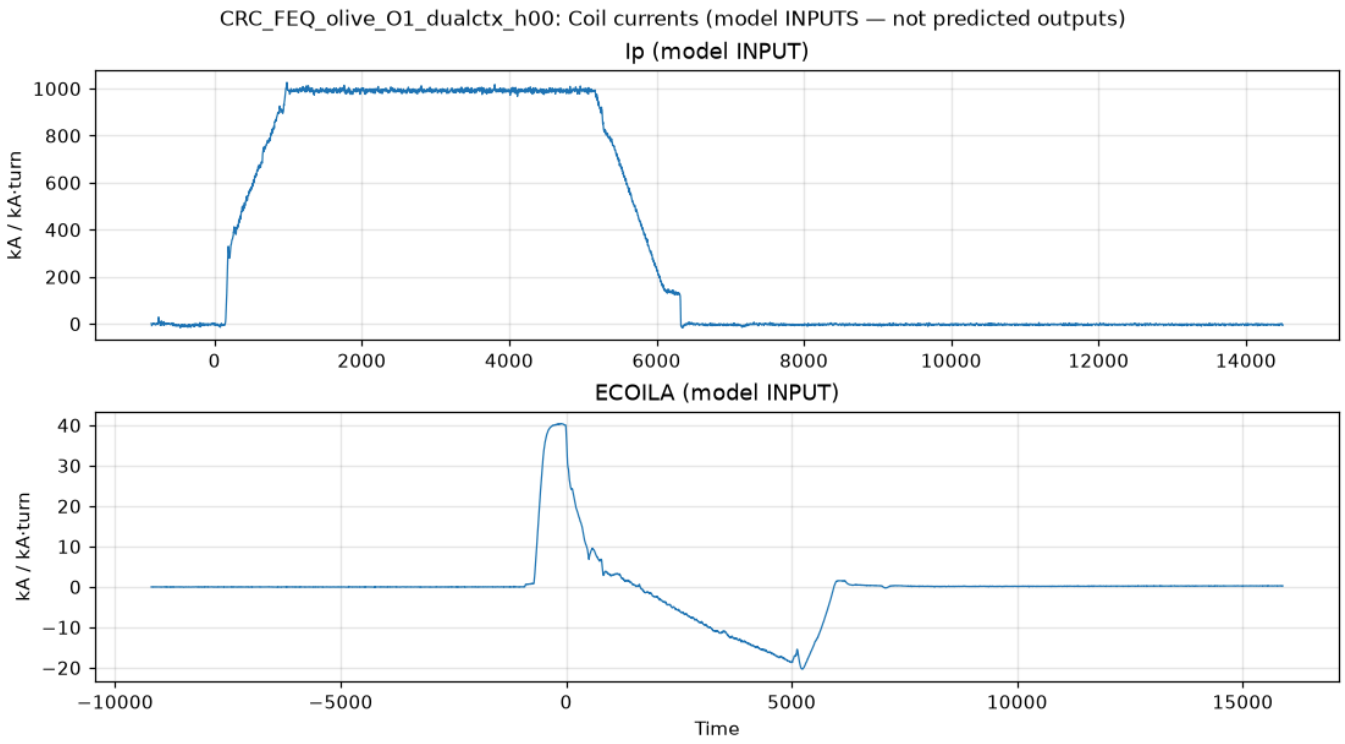
Shot 2 (`d3d_shot_bc0f6ecd42`)

- Flux MSE is low (3.10e-04) — pixel-level agreement looks strong on this shot.
- Error is core-weighted (edge/core MAE ratio 0.68) — axis / peaking may be off more than the boundary.
- Amplitude scale factor ≈ 0.992 — overall gain looks reasonable.
- Largest |Diff| at mid-time near R=2.27 m, Z=-0.05 m (|err|=0.0468).
- q95: RMSE=0.264 vs EFIT — the side-model that predicts this submitted number (separate from the 65×65 flux path) is missing the target; that hurts the 0.15 weight on $R^2_{q95,\beta N}$ in S.
- betaN: RMSE=0.0661 vs EFIT — the side-model that predicts this submitted number (separate from the flux-map path) matches well on this shot.

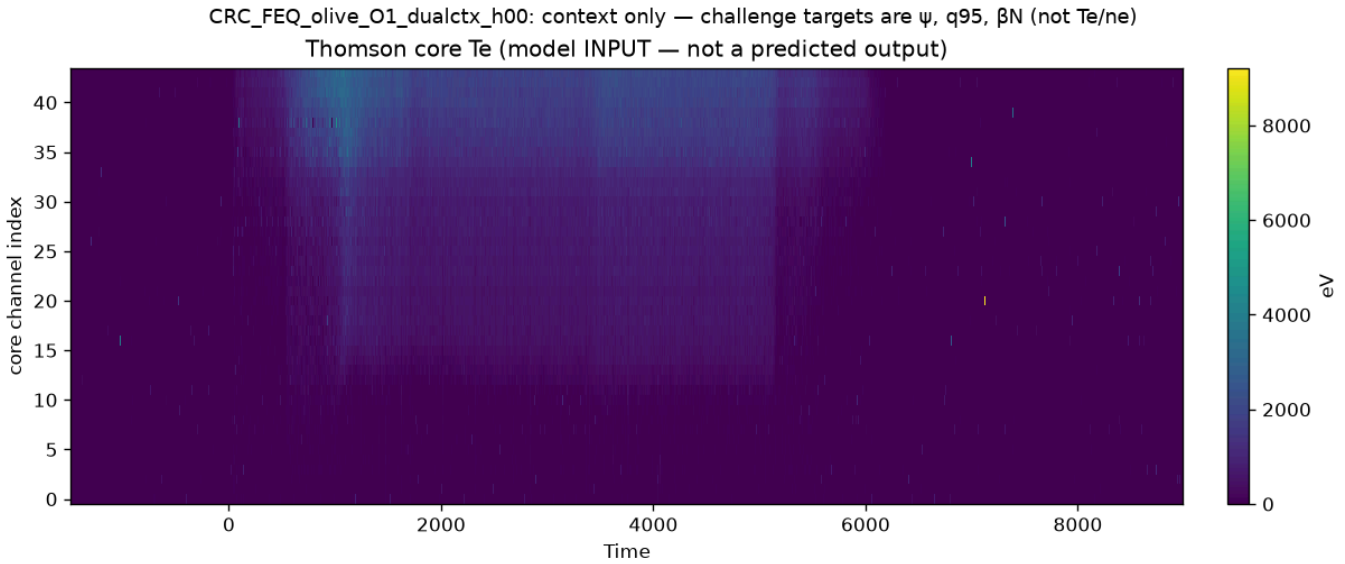
Figure pack - CRC_FEQ_olive_O1_dualctx_h00 - Shot 00

Inputs (used at inference / context — not predicted; one panel each)

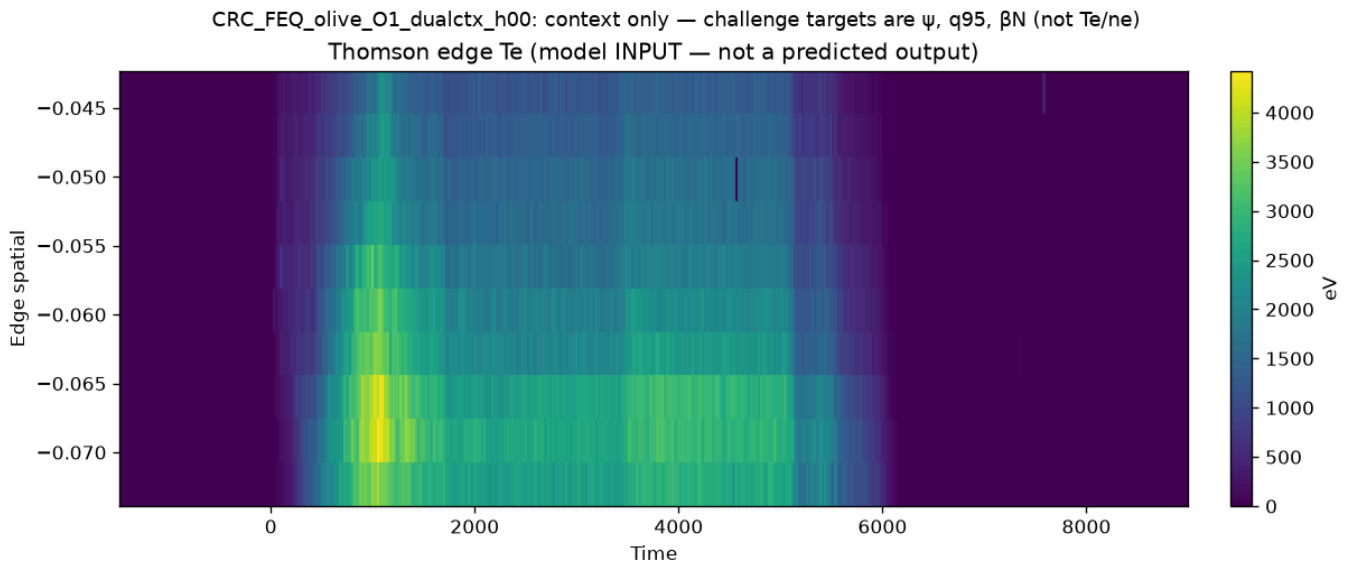
Coil currents (inputs). $I_p(t)$ = plasma current; ECOILA = ohmic solenoid. Part of the non-magnetic drive we use. Not predicted.



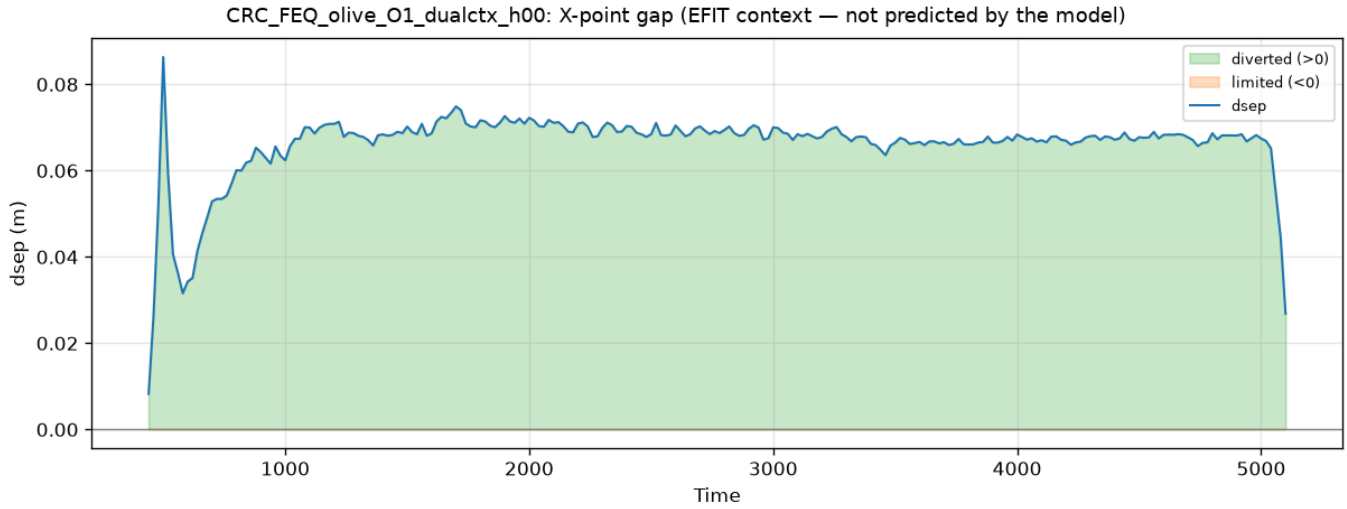
Thomson core T_e (input). Electron temperature vs channel × time from the core laser system. Kinetic profile we feed in — not a model output.



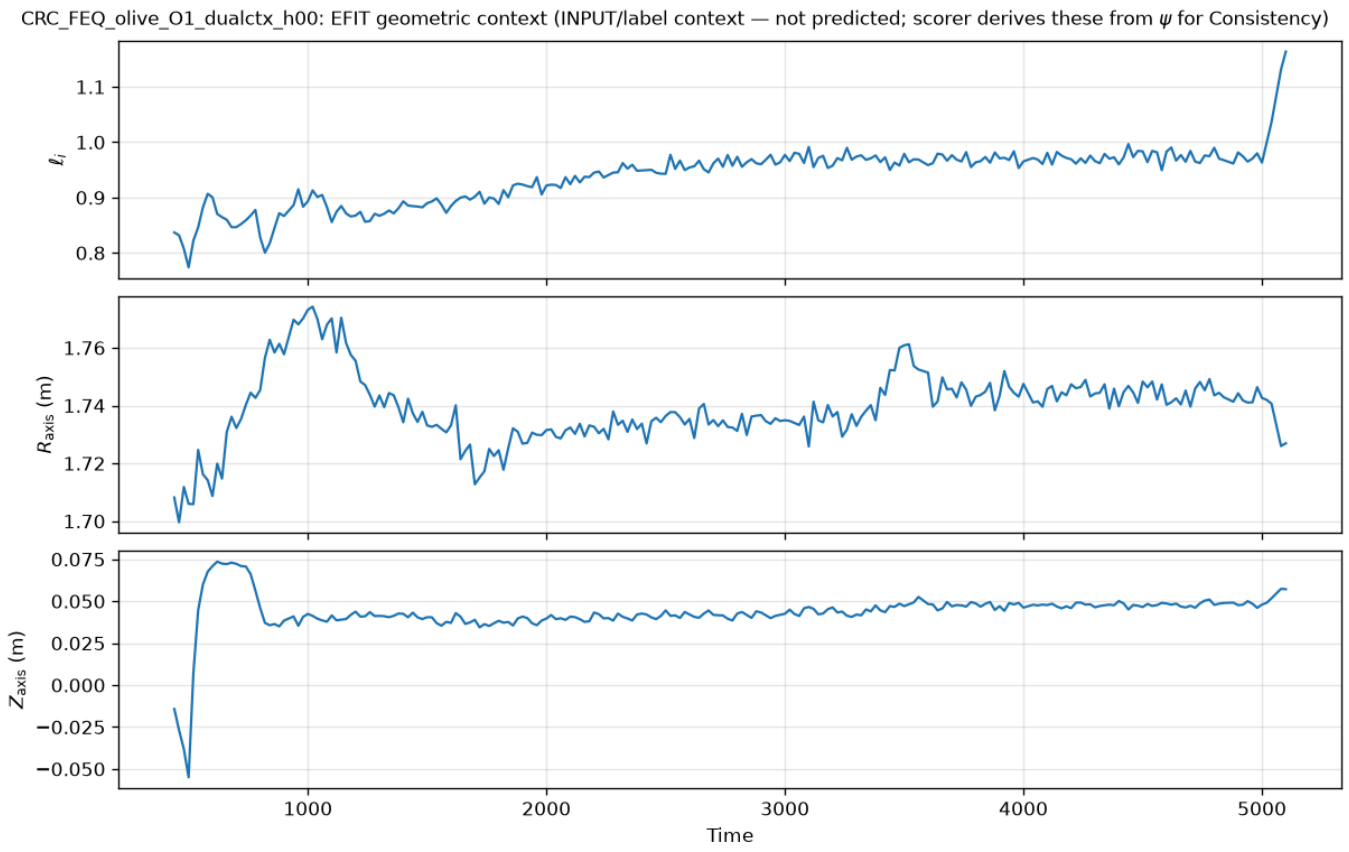
Thomson edge/tangential T_e (input). Edge-view electron temperature profile. Kinetic input we feed in — not a model output.



dsep / X-point gap (EFIT context). Signed gap (m): > 0 diverted, < 0 limited. Label for intuition; we neither submit nor predict this.



EFIT geometric context (ℓ_i, R_{axis}, Z_{axis}). Present on train labels; Consistency re-derives these from ψ. We do not submit or train heads for them here — one trace each, no Pred/Diff.

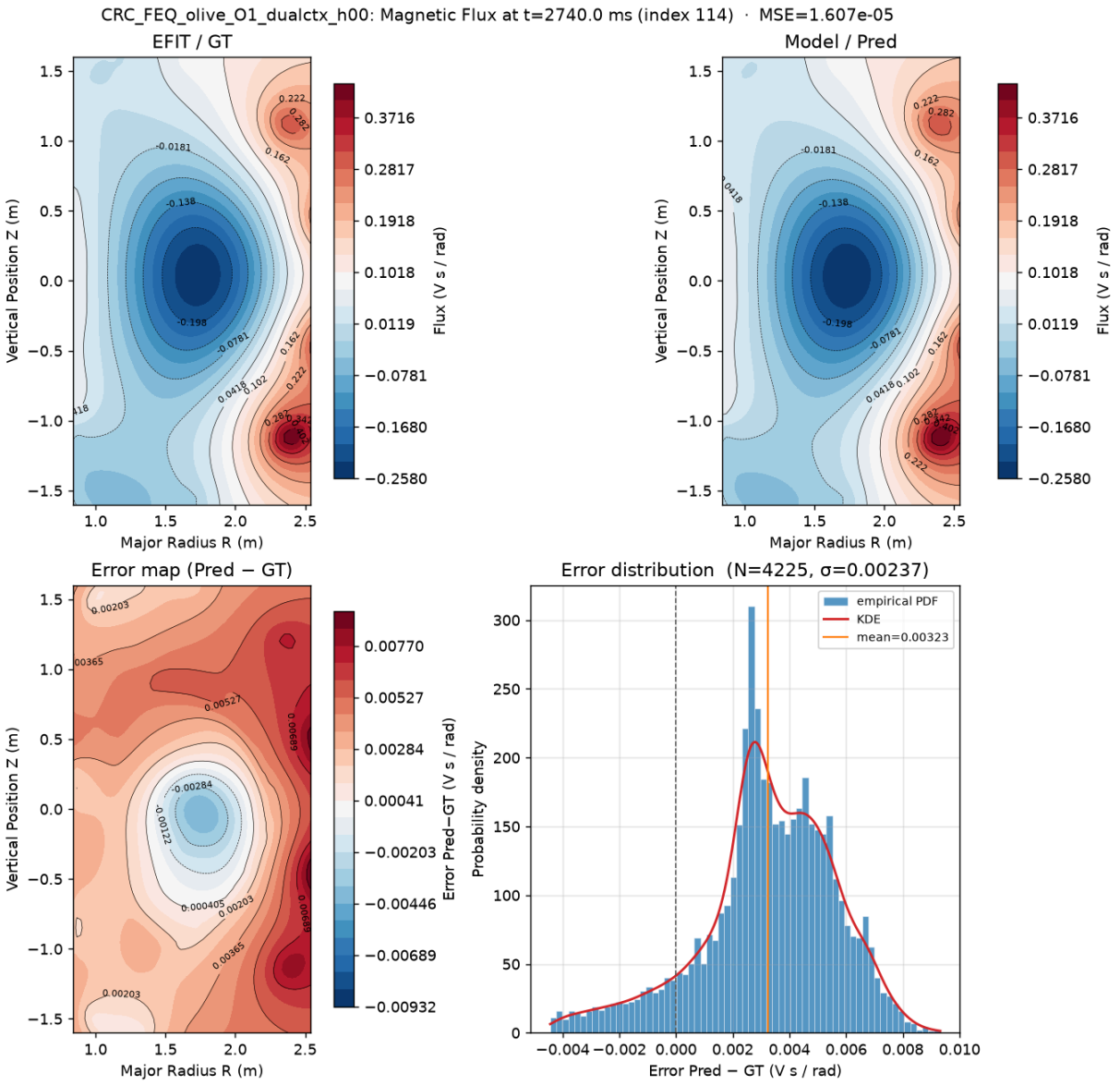


Outputs (model predictions — GT | Pred | Diff)

Flux map at mid-shot (2×2: GT, Pred, error map, error distribution).

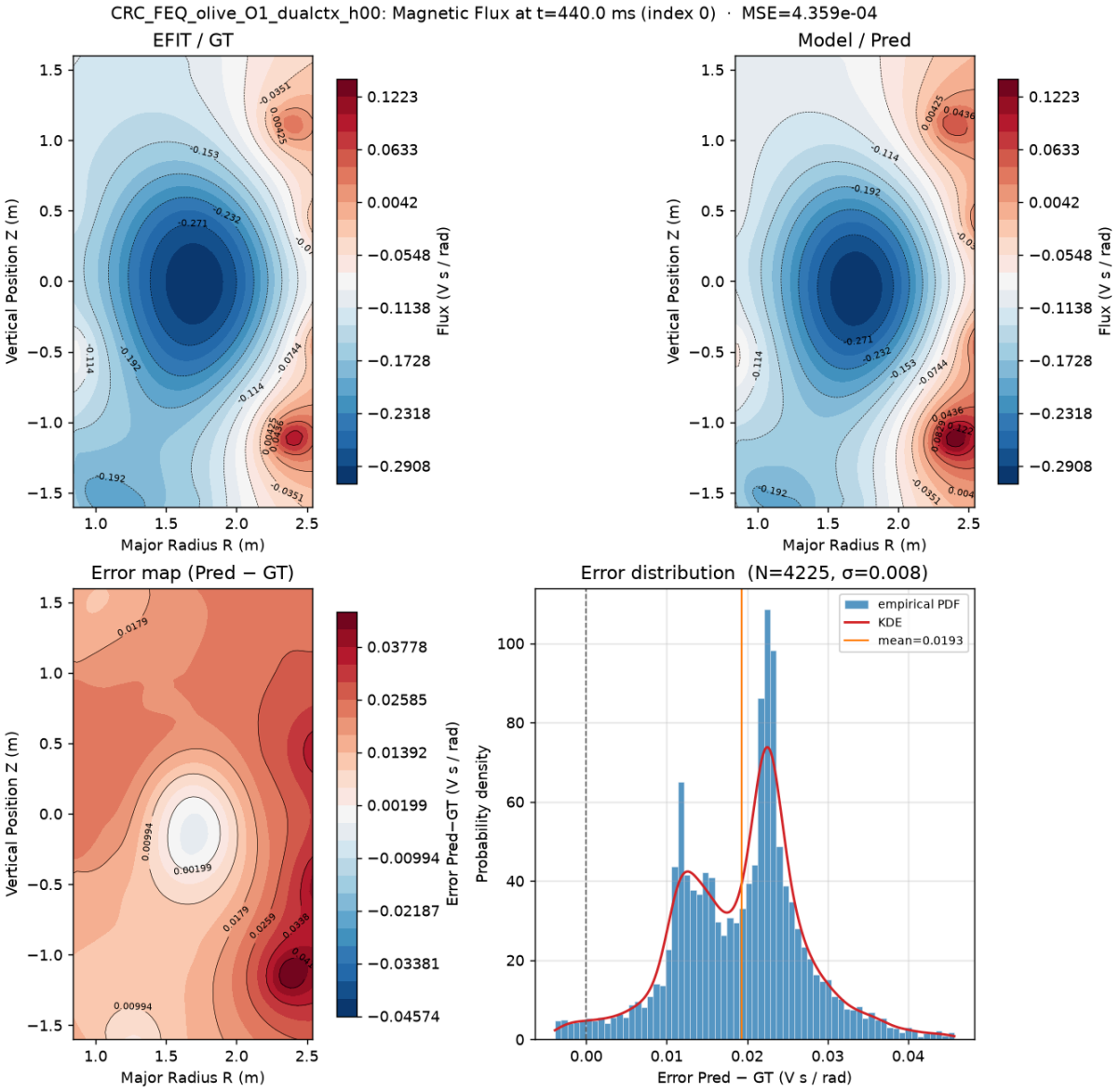
This is a vertical slice through the tokamak donut: left–right is distance from the machine centerline (major radius R), up–down is height (Z). Colors and contour lines show **poloidal magnetic flux** ψ — think of it as a topographic map of the magnetic "terrain" that cages the plasma. Nested closed loops are nested magnetic surfaces; the outer closed loop is roughly the plasma edge.

Top-left: EFIT truth. **Top-right:** our prediction. **Bottom-left:** spatial error map (Pred − GT) — where on the cross-section we miss. **Bottom-right:** empirical probability density of *all* error pixels (histogram + KDE): is the error a tight blob around zero (good) or fat-tailed / biased (systematic miss)? Mid-shot = a typical time halfway through the pulse.



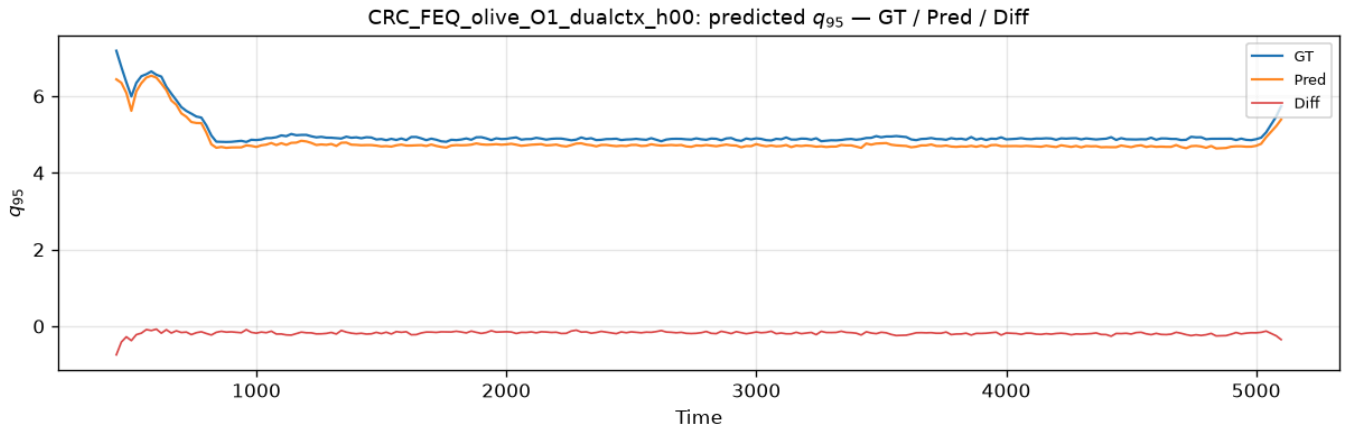
Flux map at our worst time (2×2: GT, Pred, error map, error distribution).

Same layout as the mid-shot figure, but at the EFIT time where our pixel MSE is largest. Shows *when* we fail hardest — often near start/end of the pulse or when the shape is changing quickly. The bottom-right distribution shows whether that worst frame's errors are still centered near zero or skewed.



Predicted q95

Edge safety factor q_{95} over time (GT / Pred / Diff). How twisty the magnetic field is near the plasma edge — a stability-related “safety factor” on the ~95% flux surface. Blue = EFIT truth, orange = our prediction, red = difference. Unlike ℓ_i or the axis position, this is **not** derived from our ψ by the scorer — we predict it directly and hand it in.



Predicted betaN

Normalized beta β_N over time (GT / Pred / Diff). Plasma pressure relative to the magnetic field strength (normalized). Blue = EFIT truth, orange = our prediction, red = difference. We predict this scalar directly and hand it in with the flux map.

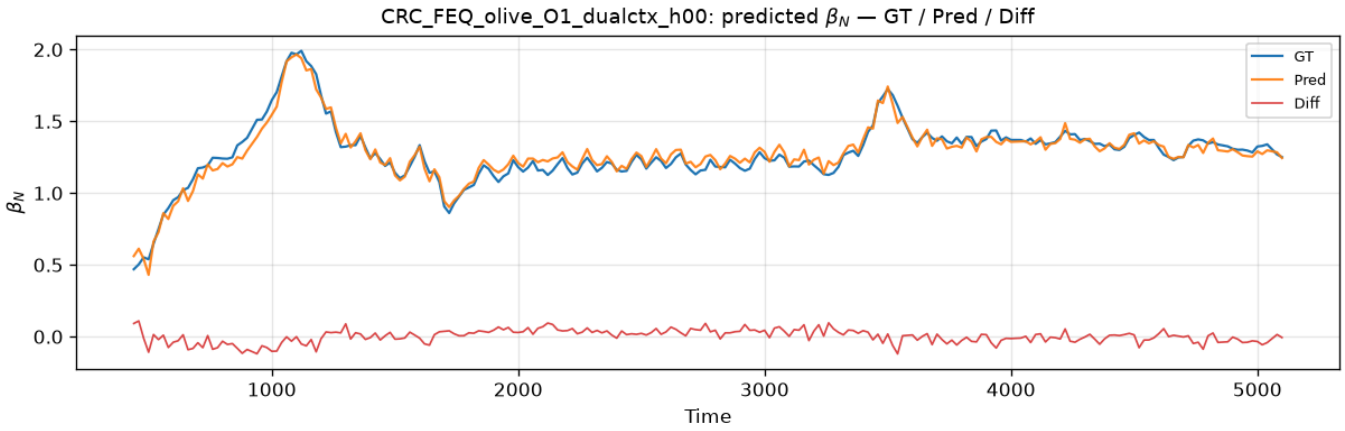
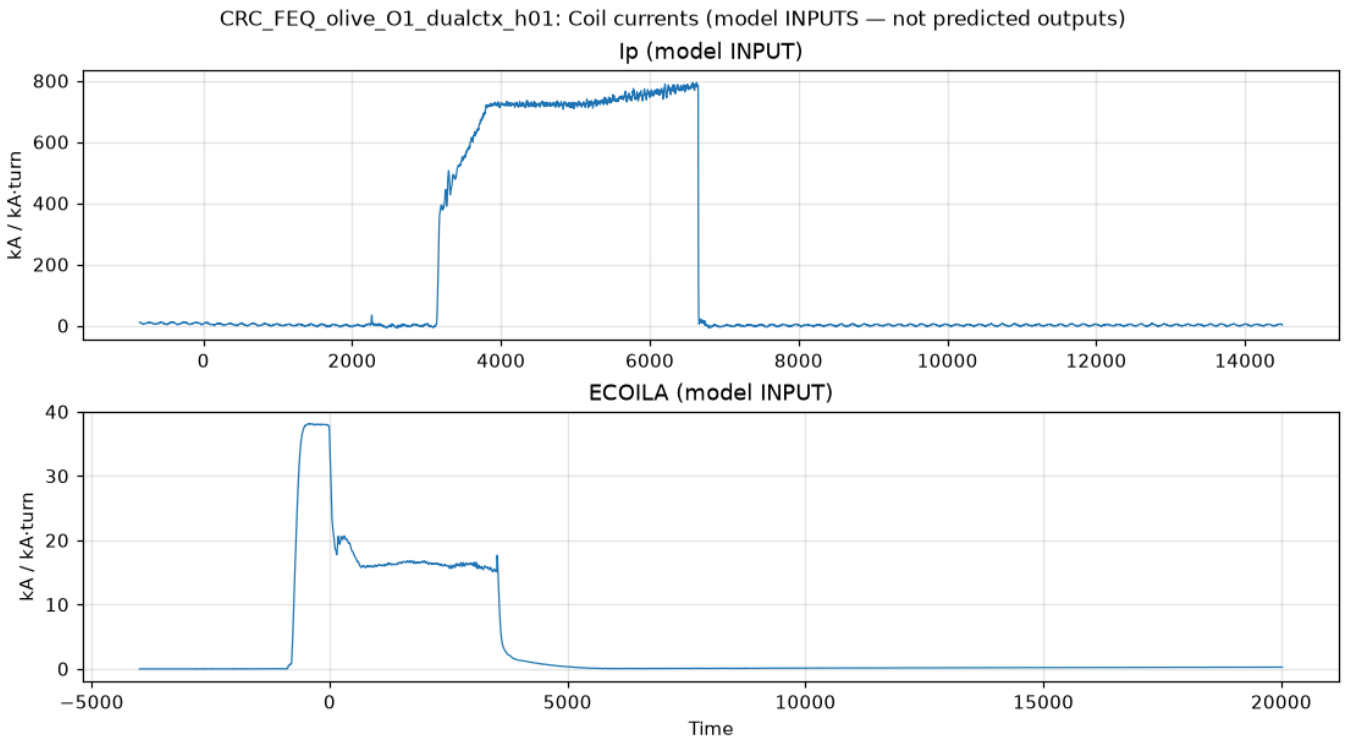


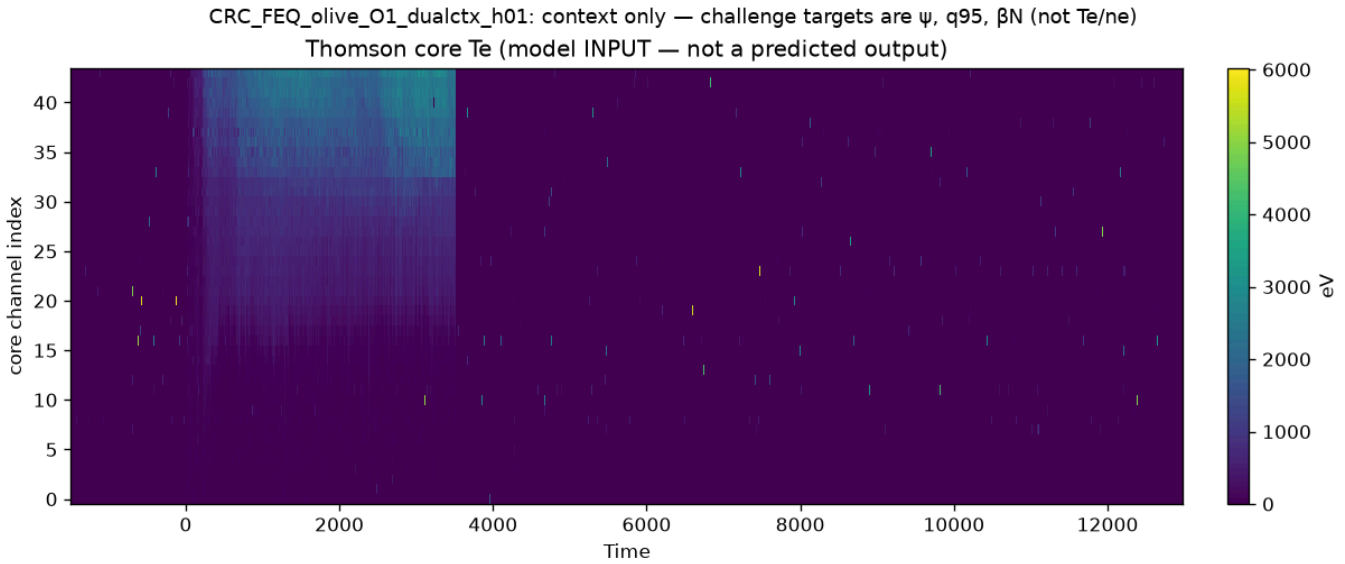
Figure pack - CRC_FEQ_olive_O1_dualctx_h01 - Shot 01

Inputs (used at inference / context — not predicted; one panel each)

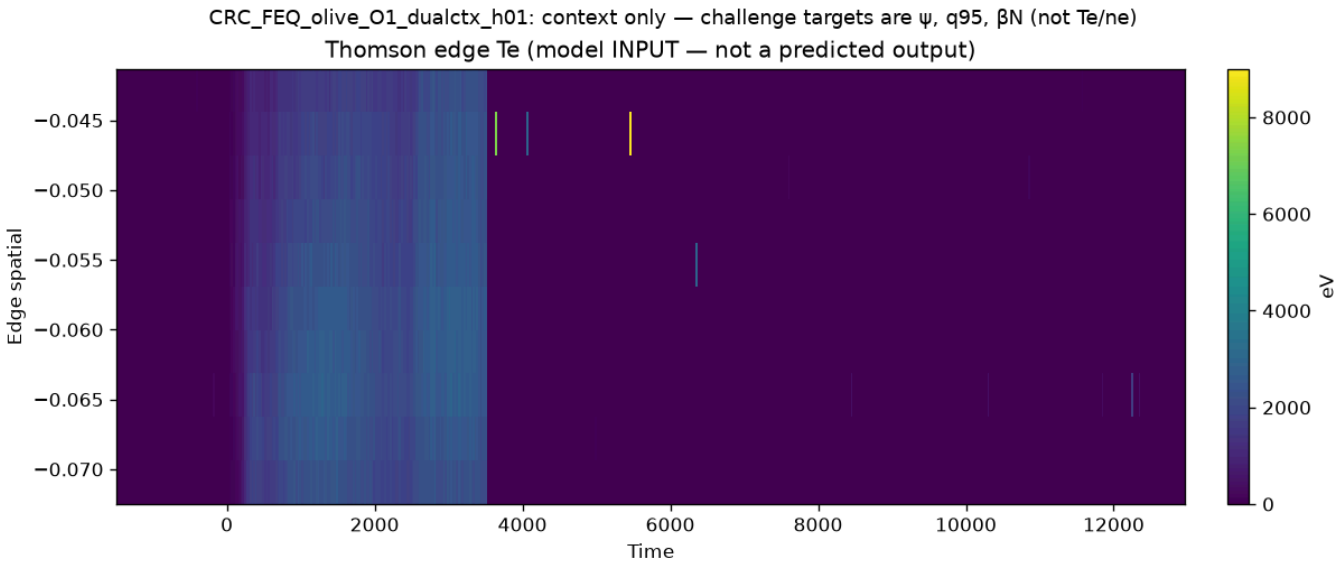
Coil currents (inputs). $I_p(t)$ = plasma current; ECOILA = ohmic solenoid. Part of the non-magnetic drive we use. Not predicted.



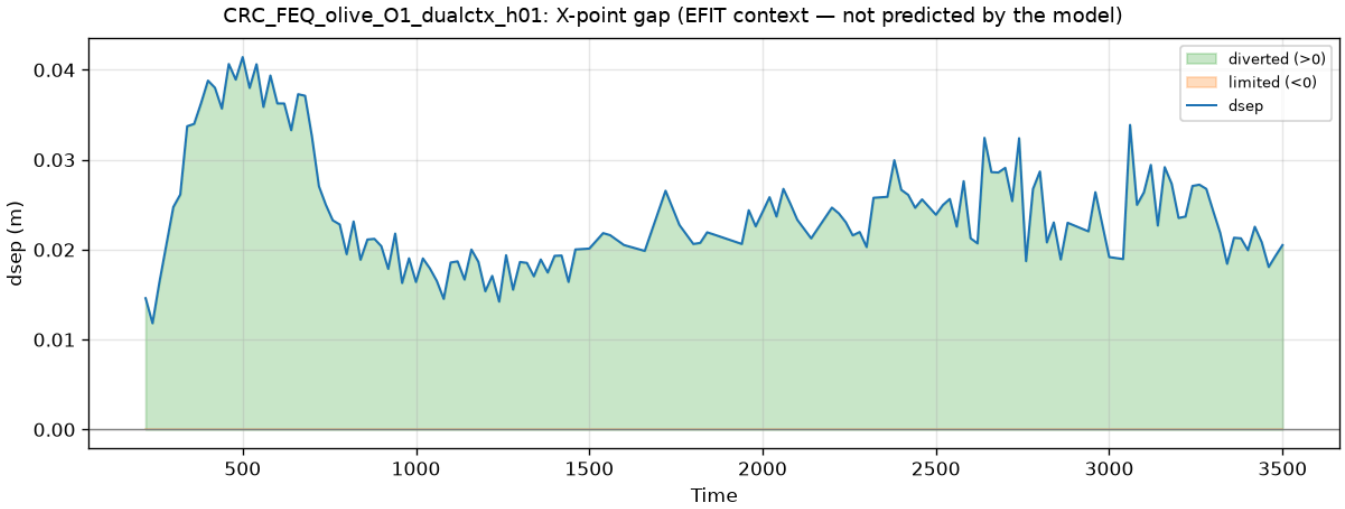
Thomson core T_e (input). Electron temperature vs channel × time from the core laser system. Kinetic profile we feed in — not a model output.



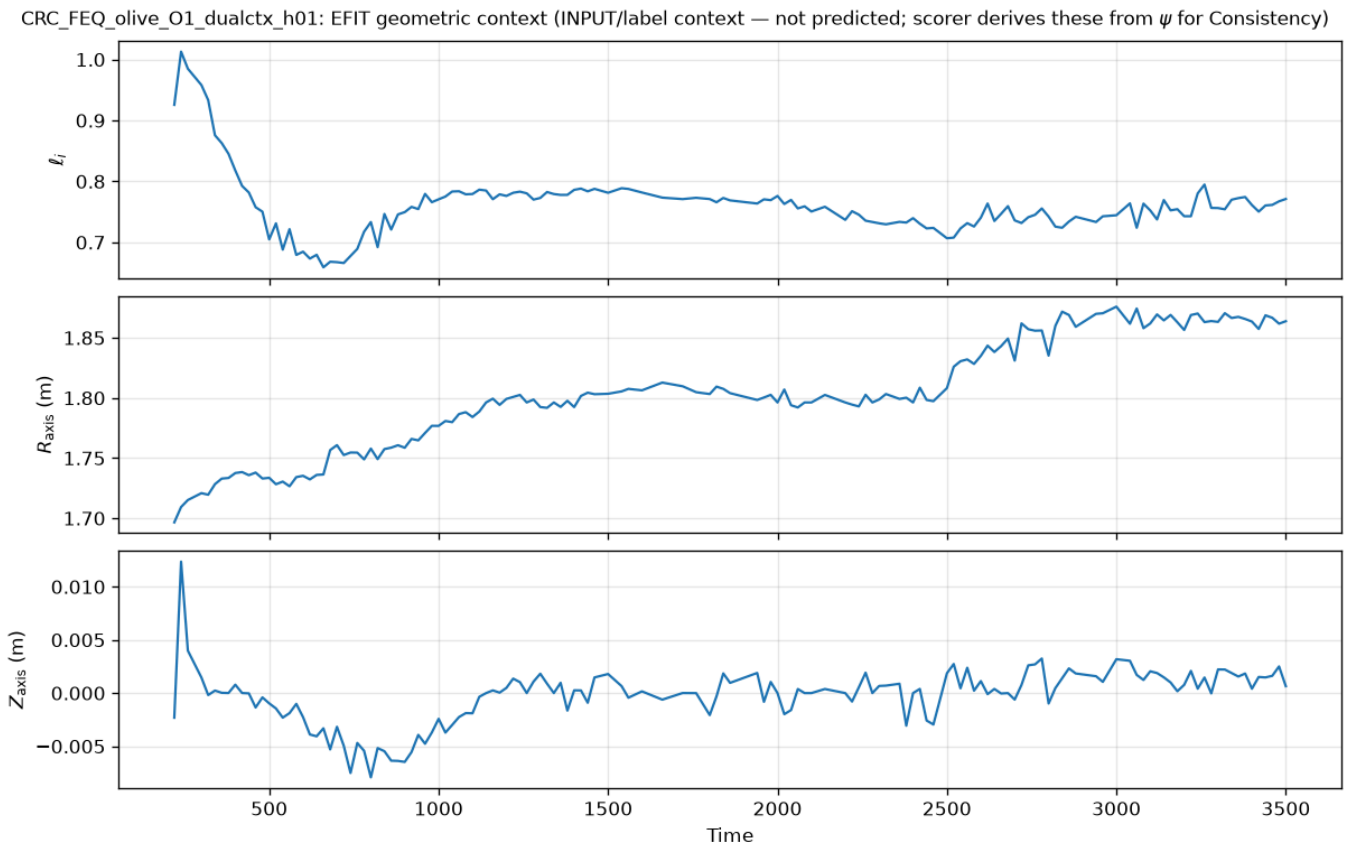
Thomson edge/tangential T_e (input). Edge-view electron temperature profile. Kinetic input we feed in — not a model output.



dsep / X-point gap (EFIT context). Signed gap (m): > 0 diverted, < 0 limited. Label for intuition; we neither submit nor predict this.



EFIT geometric context (ℓ_i, R_{axis}, Z_{axis}). Present on train labels; Consistency re-derives these from ψ. We do not submit or train heads for them here — one trace each, no Pred/Diff.

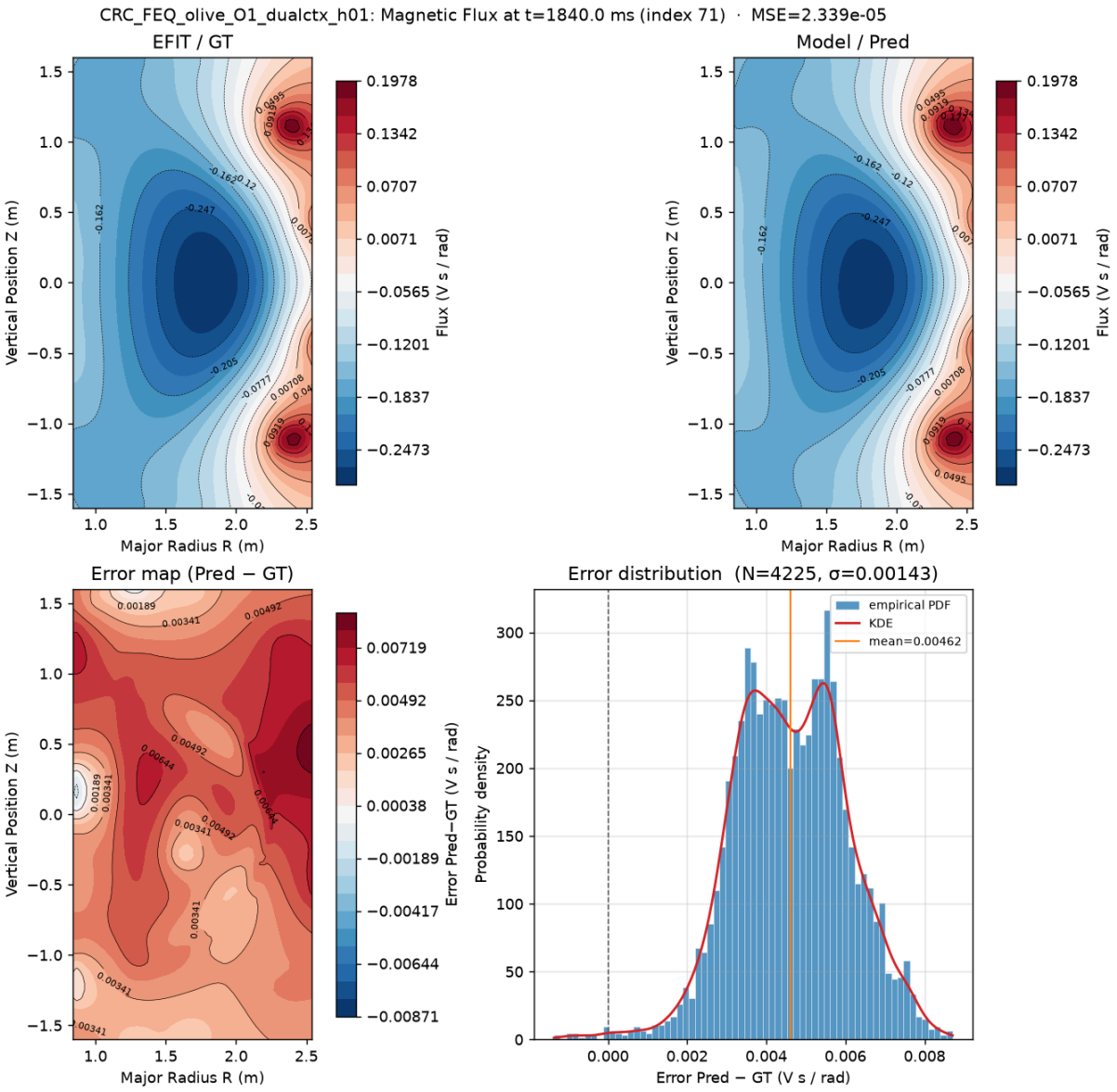


Outputs (model predictions — GT | Pred | Diff)

Flux map at mid-shot (2×2: GT, Pred, error map, error distribution).

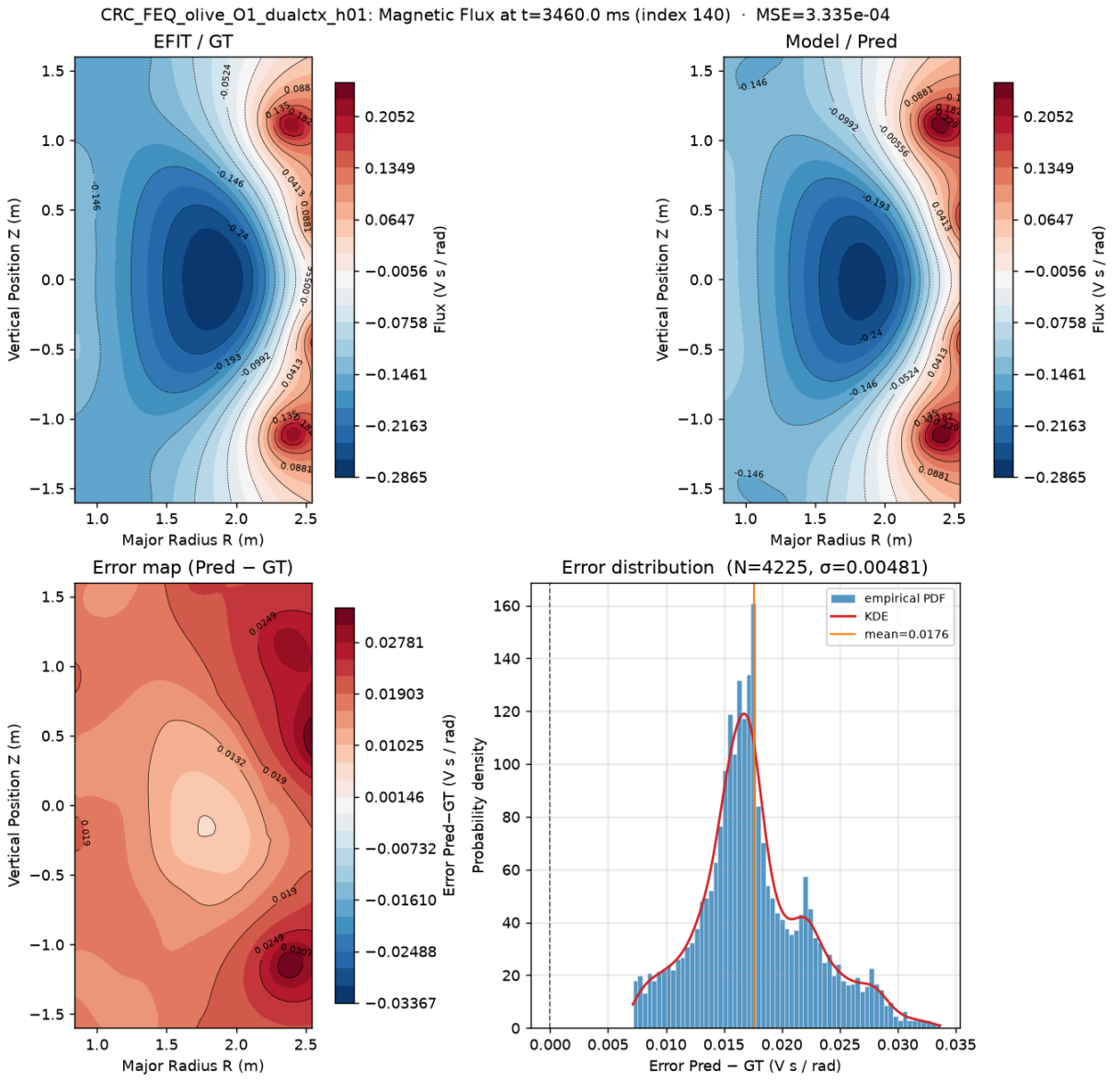
This is a vertical slice through the tokamak donut: left–right is distance from the machine centerline (major radius R), up–down is height (Z). Colors and contour lines show **poloidal magnetic flux** ψ — think of it as a topographic map of the magnetic "terrain" that cages the plasma. Nested closed loops are nested magnetic surfaces; the outer closed loop is roughly the plasma edge.

Top-left: EFIT truth. **Top-right:** our prediction. **Bottom-left:** spatial error map (Pred − GT) — where on the cross-section we miss. **Bottom-right:** empirical probability density of *all* error pixels (histogram + KDE): is the error a tight blob around zero (good) or fat-tailed / biased (systematic miss)? Mid-shot = a typical time halfway through the pulse.



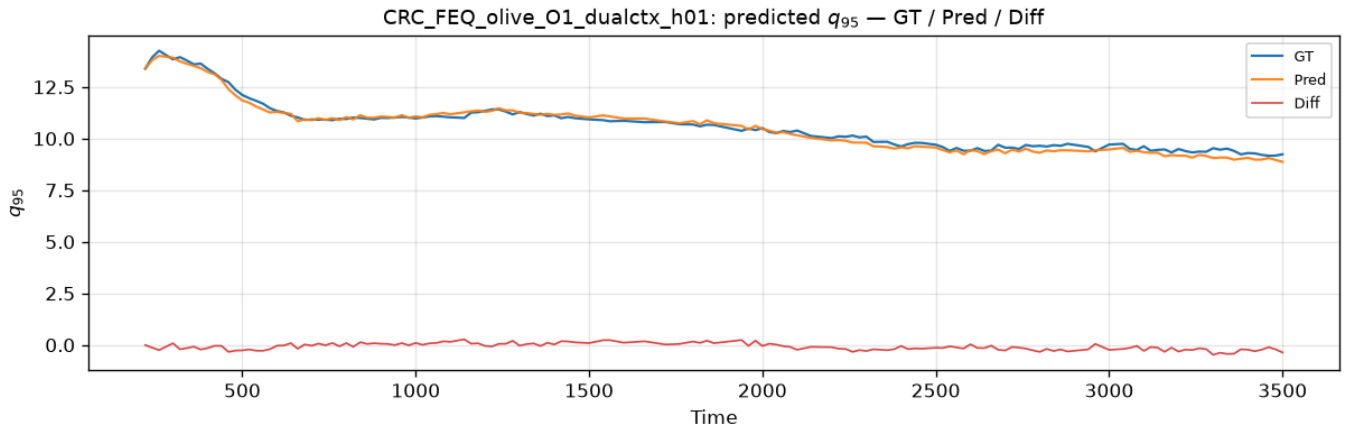
Flux map at our worst time (2×2: GT, Pred, error map, error distribution).

Same layout as the mid-shot figure, but at the EFIT time where our pixel MSE is largest. Shows *when* we fail hardest — often near start/end of the pulse or when the shape is changing quickly. The bottom-right distribution shows whether that worst frame's errors are still centered near zero or skewed.



Predicted q95

Edge safety factor q_{95} over time (GT / Pred / Diff). How twisty the magnetic field is near the plasma edge — a stability-related “safety factor” on the ~95% flux surface. Blue = EFIT truth, orange = our prediction, red = difference. Unlike ℓ_i or the axis position, this is **not** derived from our ψ by the scorer — we predict it directly and hand it in.



Predicted betaN

Normalized beta β_N over time (GT / Pred / Diff). Plasma pressure relative to the magnetic field strength (normalized). Blue = EFIT truth, orange = our prediction, red = difference. We predict this scalar directly and hand it in with the flux map.

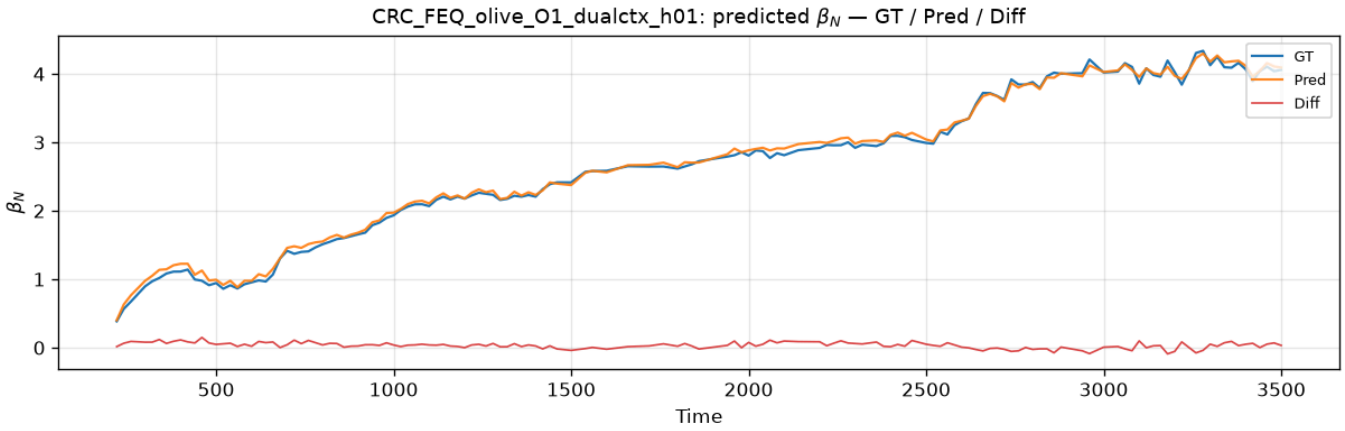
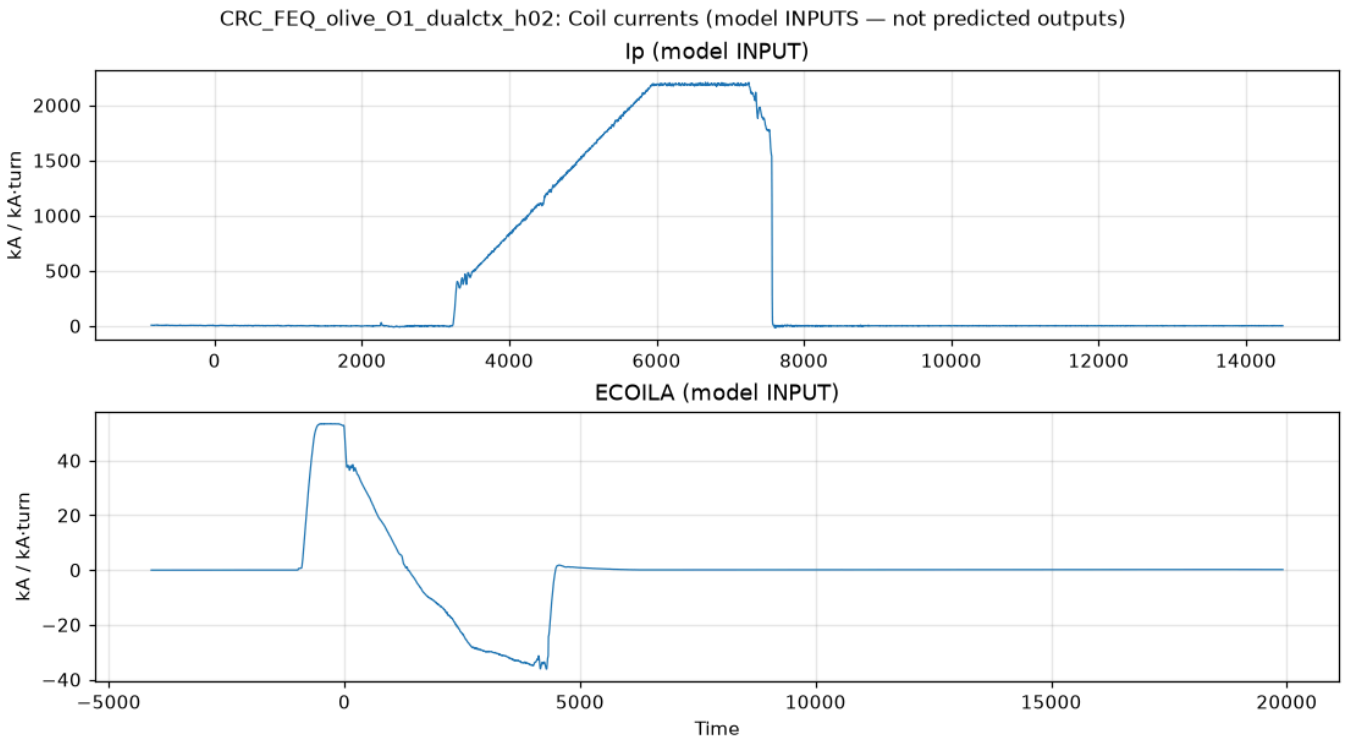


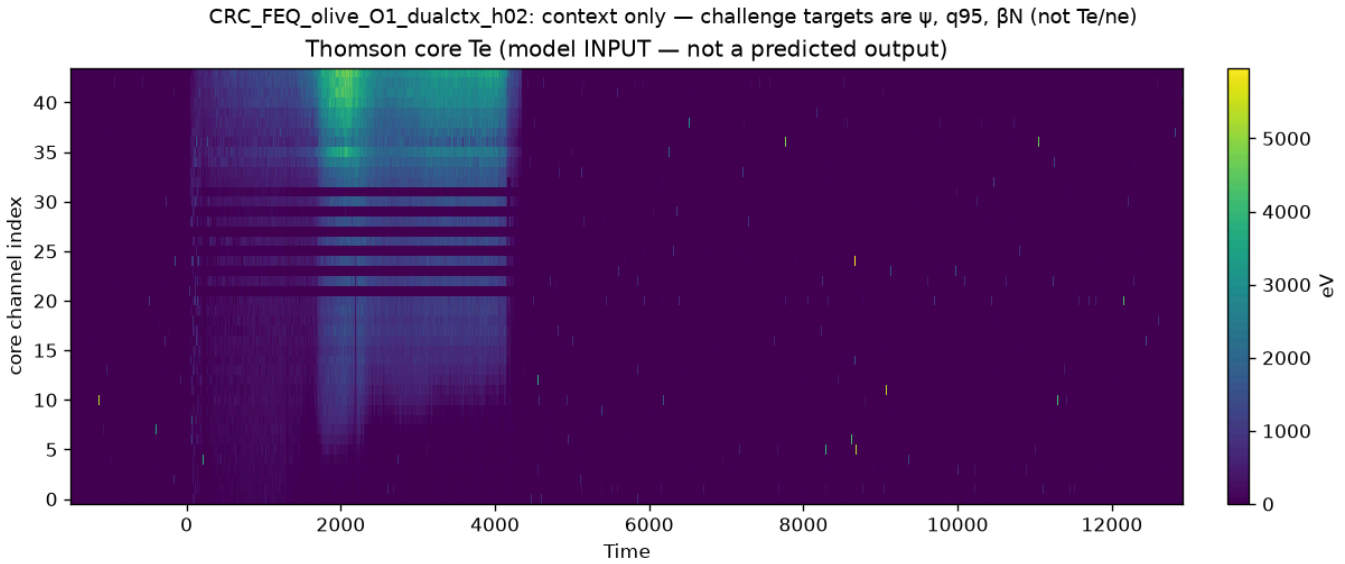
Figure pack - CRC_FEQ_olive_O1_dualctx_h02 - Shot 02

Inputs (used at inference / context — not predicted; one panel each)

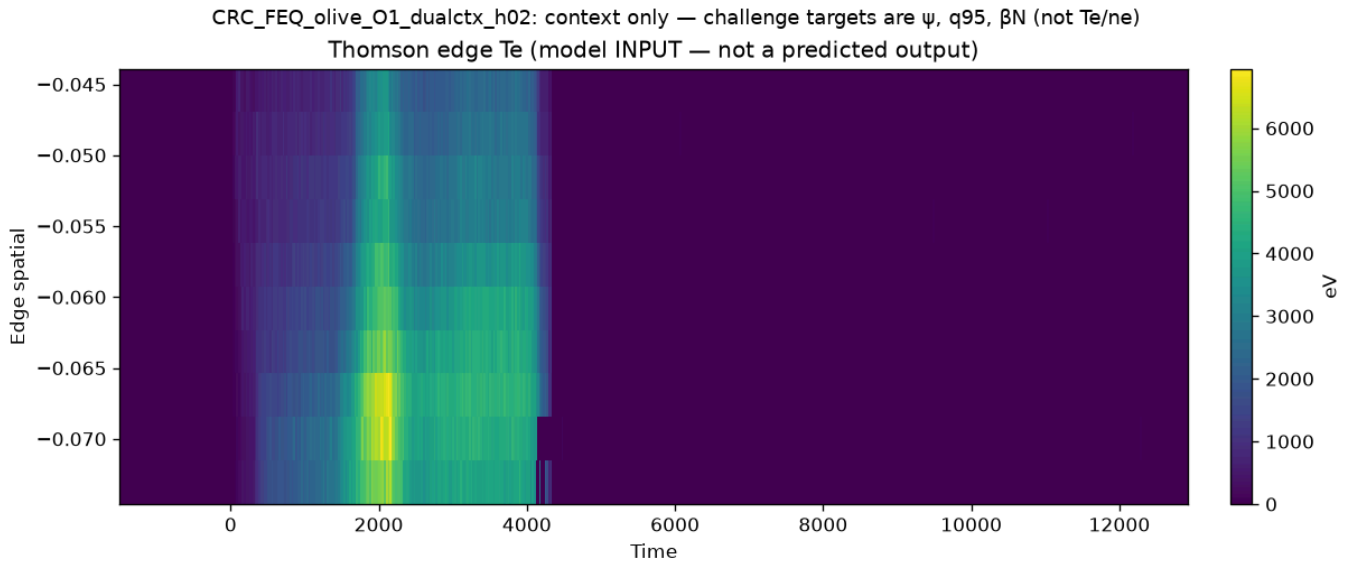
Coil currents (inputs). $I_p(t)$ = plasma current; ECOILA = ohmic solenoid. Part of the non-magnetic drive we use. Not predicted.



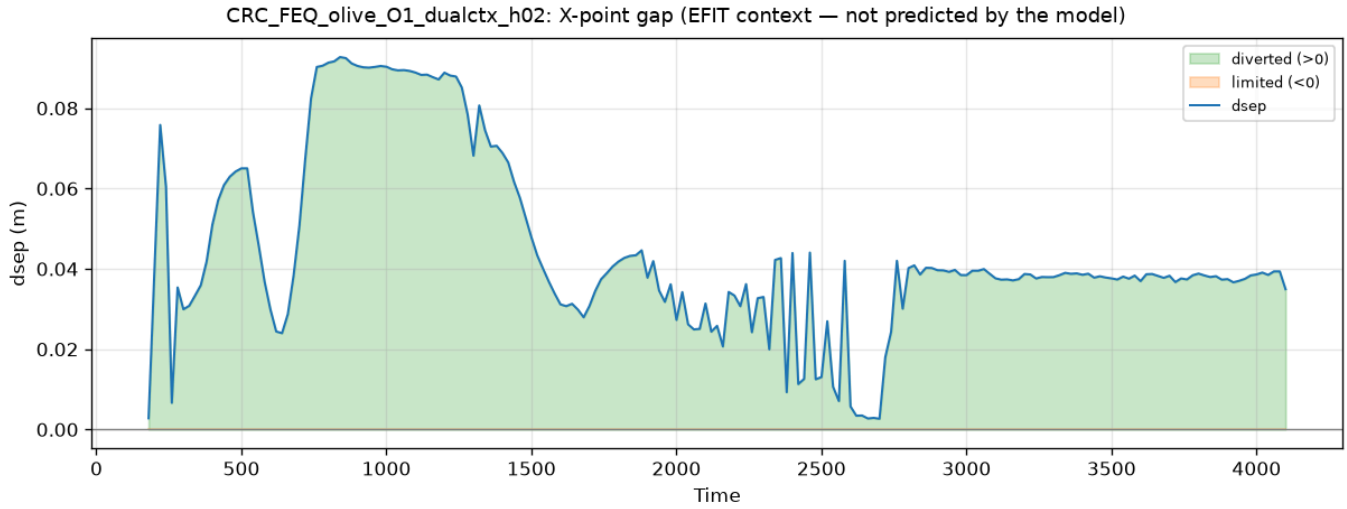
Thomson core T_e (input). Electron temperature vs channel × time from the core laser system. Kinetic profile we feed in — not a model output.



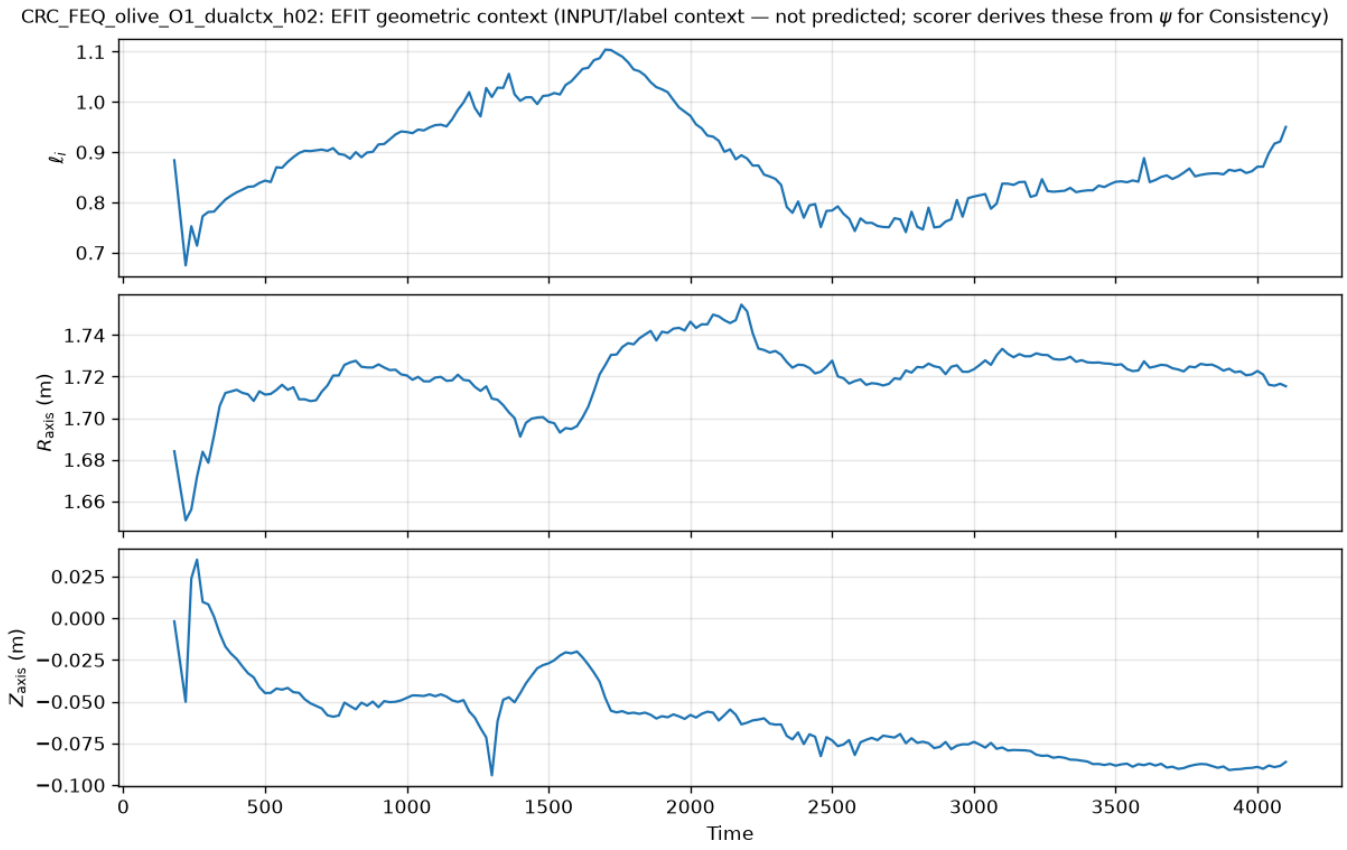
Thomson edge/tangential T_e (input). Edge-view electron temperature profile. Kinetic input we feed in — not a model output.



dsep / X-point gap (EFIT context). Signed gap (m): > 0 diverted, < 0 limited. Label for intuition; we neither submit nor predict this.



EFIT geometric context (ℓ_i, R_{axis}, Z_{axis}). Present on train labels; Consistency re-derives these from ψ. We do not submit or train heads for them here — one trace each, no Pred/Diff.

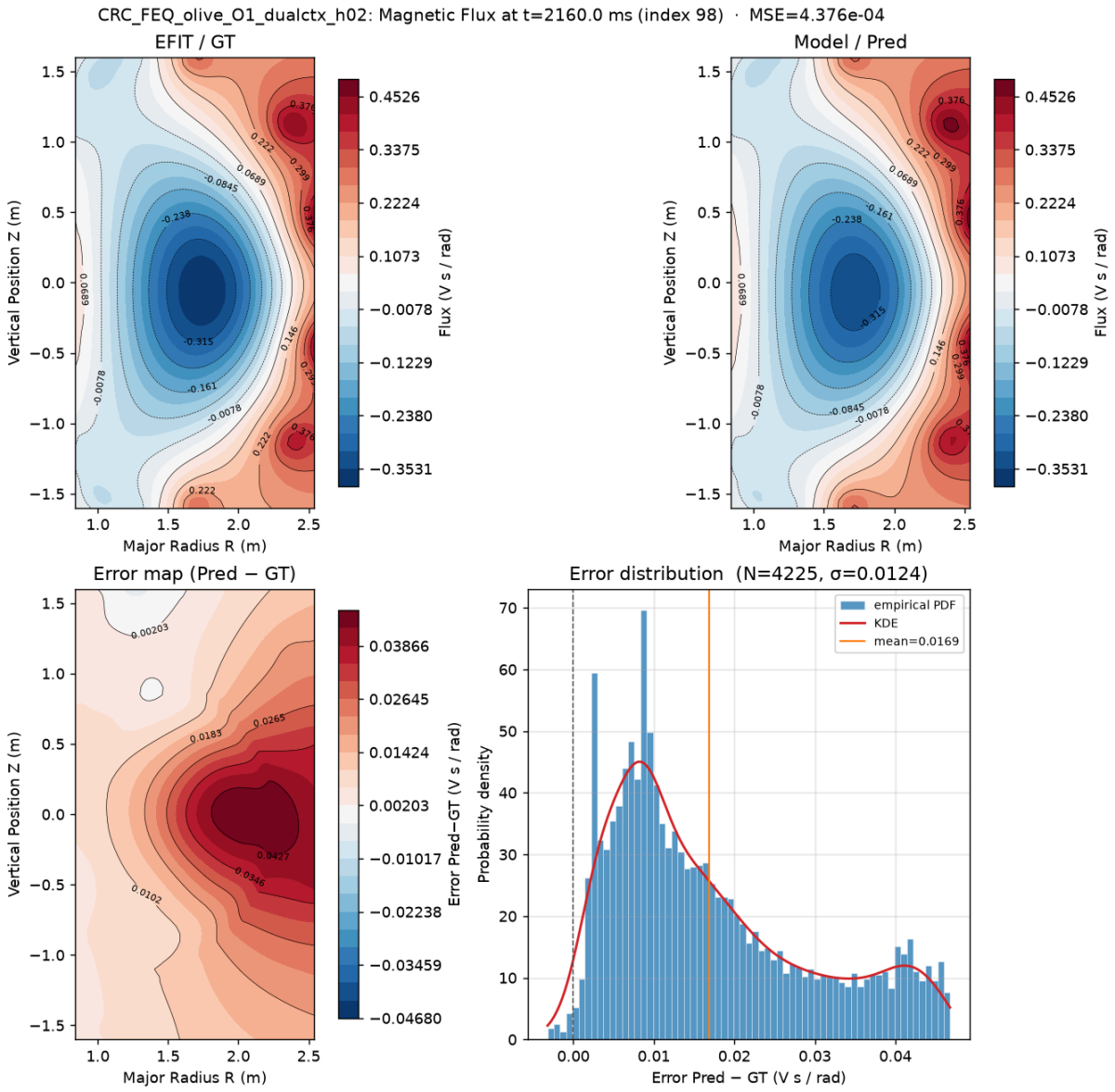


Outputs (model predictions — GT | Pred | Diff)

Flux map at mid-shot (2×2: GT, Pred, error map, error distribution).

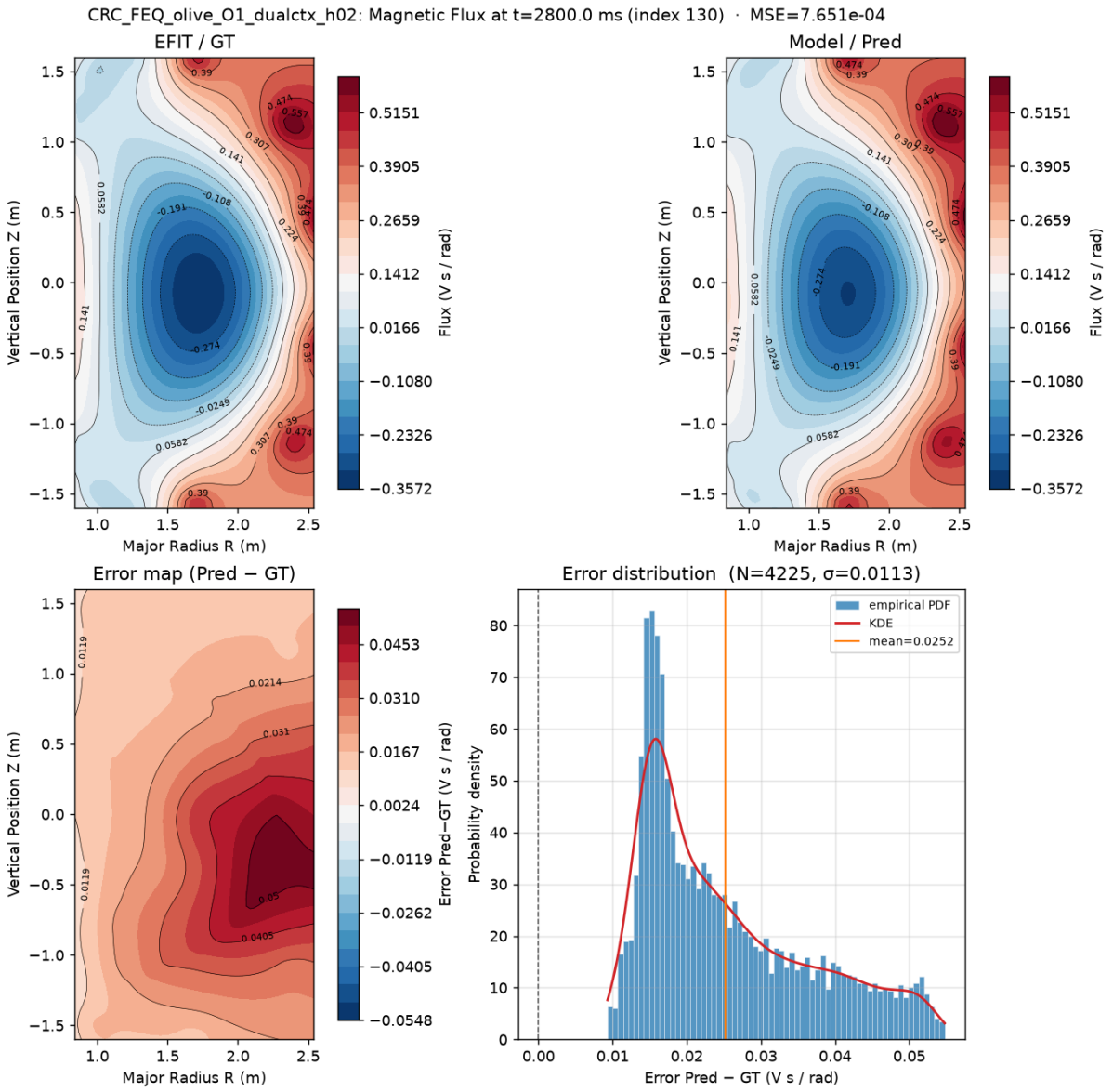
This is a vertical slice through the tokamak donut: left–right is distance from the machine centerline (major radius R), up–down is height (Z). Colors and contour lines show **poloidal magnetic flux** ψ — think of it as a topographic map of the magnetic "terrain" that cages the plasma. Nested closed loops are nested magnetic surfaces; the outer closed loop is roughly the plasma edge.

Top-left: EFIT truth. **Top-right:** our prediction. **Bottom-left:** spatial error map (Pred − GT) — where on the cross-section we miss. **Bottom-right:** empirical probability density of *all* error pixels (histogram + KDE): is the error a tight blob around zero (good) or fat-tailed / biased (systematic miss)? Mid-shot = a typical time halfway through the pulse.



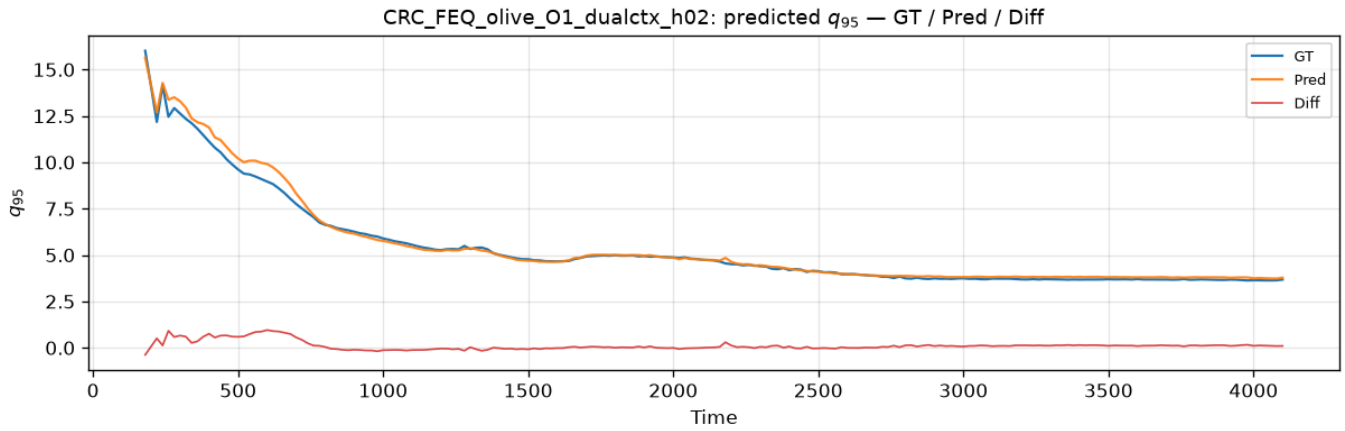
Flux map at our worst time (2×2: GT, Pred, error map, error distribution).

Same layout as the mid-shot figure, but at the EFIT time where our pixel MSE is largest. Shows *when* we fail hardest — often near start/end of the pulse or when the shape is changing quickly. The bottom-right distribution shows whether that worst frame's errors are still centered near zero or skewed.



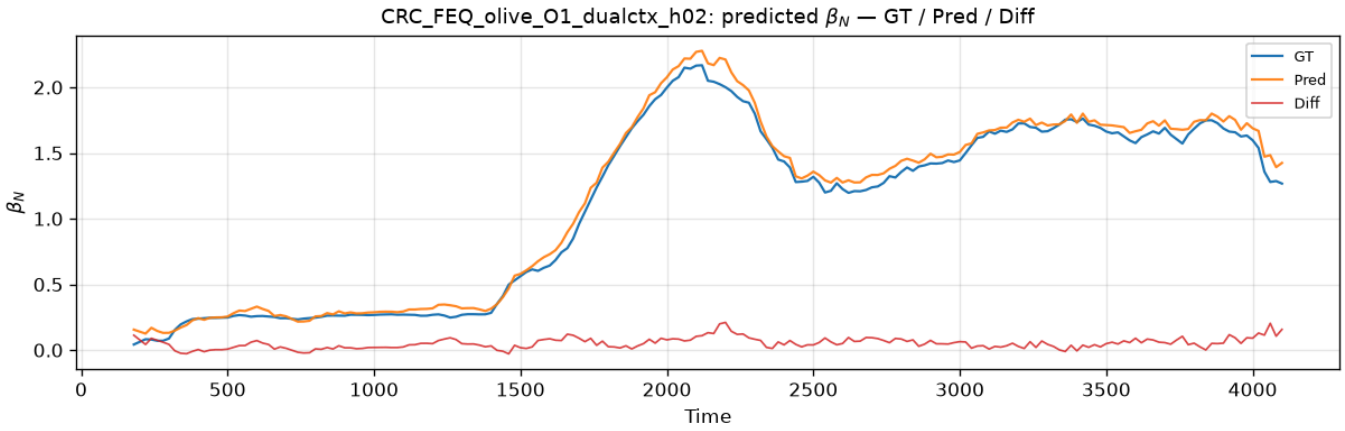
Predicted q95

Edge safety factor q_{95} over time (GT / Pred / Diff). How twisty the magnetic field is near the plasma edge — a stability-related “safety factor” on the ~95% flux surface. Blue = EFIT truth, orange = our prediction, red = difference. Unlike ℓ_i or the axis position, this is **not** derived from our ψ by the scorer — we predict it directly and hand it in.



Predicted betaN

Normalized beta β_N over time (GT / Pred / Diff). Plasma pressure relative to the magnetic field strength (normalized). Blue = EFIT truth, orange = our prediction, red = difference. We predict this scalar directly and hand it in with the flux map.



Notes

- Authoritative train-era model narrative: `catskills/MODEL_DESCRIPTION_CRC_FEQ_olive.md` (status line there still says campaign CLOSED; this REPORT is the checkpoint-centric update).
- Authoritative afterlife / mixes: `catskills/COLOR_LEDGER.md` §30.10–§30.11.
- Organizer rebuild pointer: `ORGANIZER_REPLY_REBUILD.md`.
- Figure pack: `weights/phase3/figures/shot_{00,01,02}/` (first 3 holdout shots, seed 42 / 6961+80 split). Regenerated 2026-11-03.
- No `direct_submission.zip` for this weight (parent checkpoint, not a Codabench attempt).